

Configuration and provenance

Standalone EJ-230 report

Branch: feat/ej230-ssl4

Generation source commit: ebbb630434ce

Data session: session_20260612_171407 on
t0minidaq

Data path: results_ej230/data; generated
2026-06-12

Report generation: 2026-06-13 16:20 CEST

- ▶ 31 positions $\times$ 2000 events; main scan produced on t0minidaq with 24 threads.
- ▶ Directed controls on MSI with 16 threads: 4 EndTop $\times$ 2000 events and 3 End-only $\times$ 10000 events.
- ▶ 16 End + 70 Top channels; jitter fixed to zero.
- ▶ EJ-230 via SSLG4 OPSC-106: yield 9700/MeV, $\tau_r = 0.5$ ns, $\tau_d = 1.5$ ns, ABSLENGTH=120 cm.

Scope

Top 70 SiPM is a simulation coverage study. Real hardware: 32 Top SiPMs. All reported timing is intrinsic, without SPTR or electronics jitter.

EndTop geometry and channel map

- ▶ EJ-230 bar: $1400 \times 60 \times 10 \text{ mm}^3$.
- ▶ End L IDs 0–7; End R IDs 8–15.
- ▶ Top L IDs 16–50: $-692, -672, \dots, -12 \text{ mm}$.
- ▶ Top R IDs 51–85: $+12, +32, \dots, +692 \text{ mm}$.
- ▶ Each Top element and window: $6 \times 6 \text{ mm}^2$.

Exact constraints

First centers are 5 mm from each bar end. Pitch is 20 mm except the deliberate central $-12/+12 \text{ mm}$ pair, whose pitch is 24 mm.

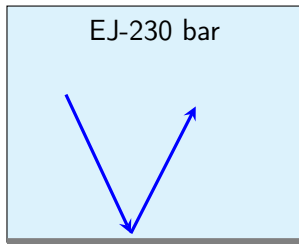
Instrumented and wrapped faces

Face	Optical state	Readout
$-X, +X$	open	8 End SiPMs each
$+Y$	Mylar with 70 exact windows	70 Top SiPMs
$-Y, -Z, +Z$	complete Mylar	none

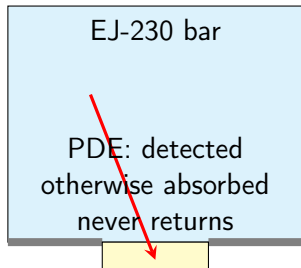
External layer

Black Tedlar is deliberately not modeled; it only blocks ambient light. The non-reflected Mylar fraction is already absorbed in the optical model.

Window optics: why timing and collection change



Mylar, fixed $R = 0.98$: reflected photons return

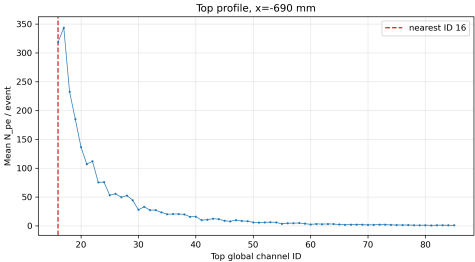
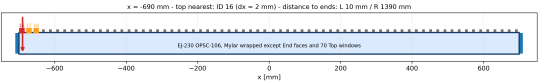


index-matched window

Consequence

Every window is an optical drain. Long, multiply reflected paths have more chances to meet one, so late End photons are preferentially removed.

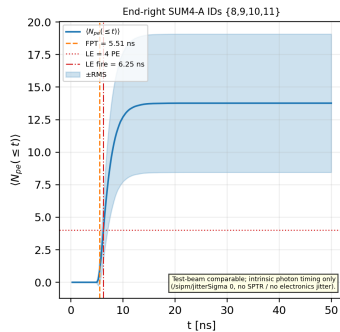
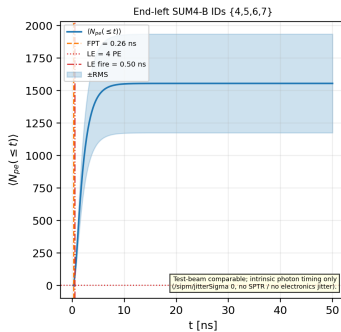
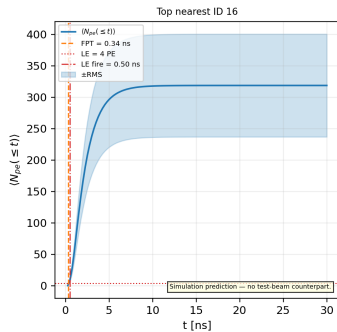
Position $x=-690$ mm



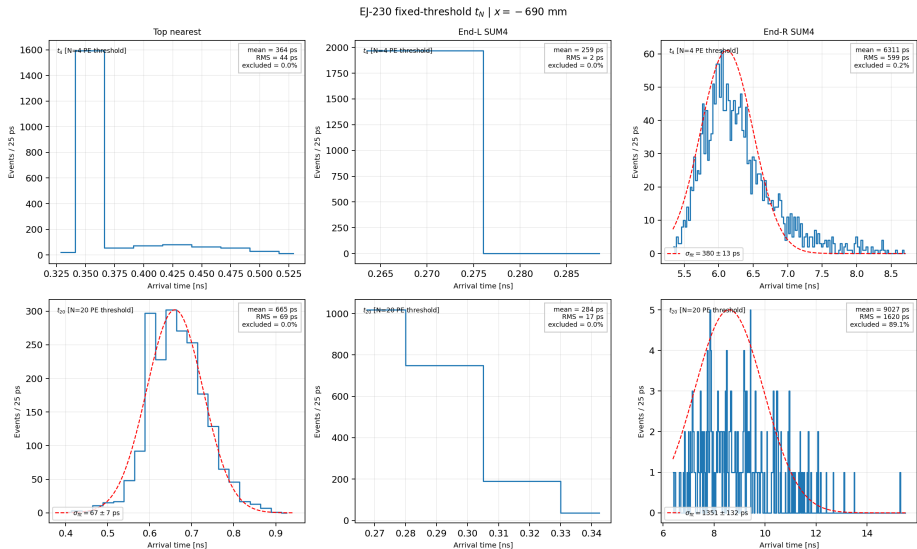
Quantity	Value
End L N_{pe}	3073.3 ± 16.8
End R N_{pe}	27.4 ± 0.2
Nearest Top ID	16 (window-track exception)
Nearest Top N_{pe}	318.7 ± 1.8
Nearest Top $\langle t \rangle$	2.3460 ± 0.0020 ns
Nearest Top FPT	0.34237 ± 0.00012 ns

Run 1 | $x = -690$ mm | Photon arrival $\langle N(t) \rangle$

$x = -690$ mm | Photon arrival $\langle N(t) \rangle$

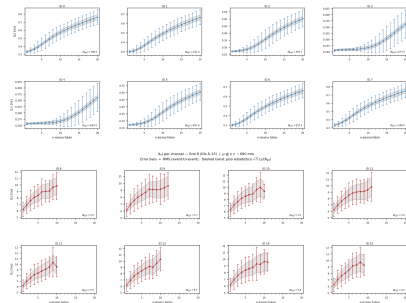
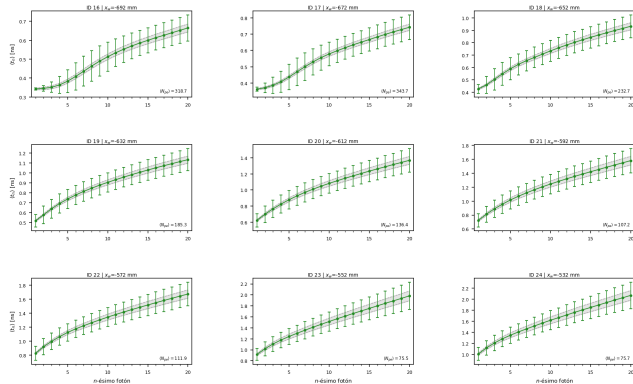


Run 1 | $x = -690$ mm | Time-to-threshold t_N



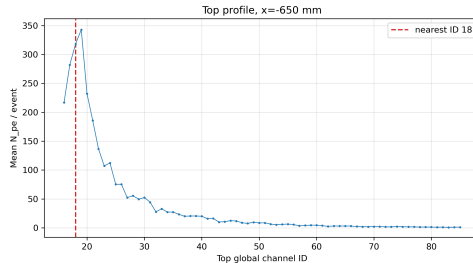
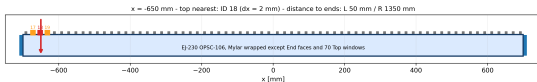
Run 1 | $x = -690$ mm | $\langle t_n \rangle$ per channel (dispersion)

$\langle t_n \rangle$ per Top channel (nearest ± 4) | $\mu @ x = -690$ mm
Error bars = RMS. Dashed band: piso estadístico $\sqrt{n} \tau_d / \langle N_{pe} \rangle$



Error bars = RMS (event-to-event). Dashed band: statistical floor $\sigma_{stat}(t_n) \approx \sqrt{n} \tau_d / \langle N_{pe} \rangle$ ($\tau_d = 1.5$ ns). See dispersion-decomposition frame for physical interpretation.

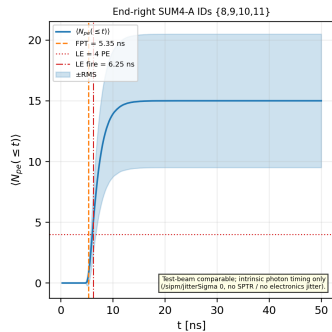
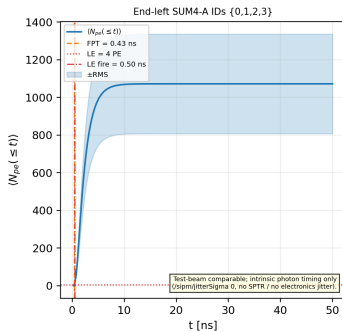
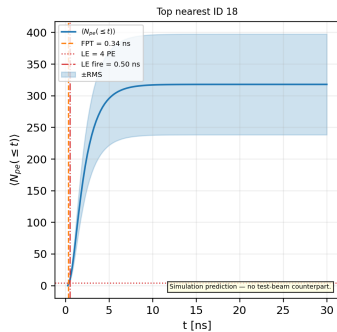
Position $x=-650$ mm



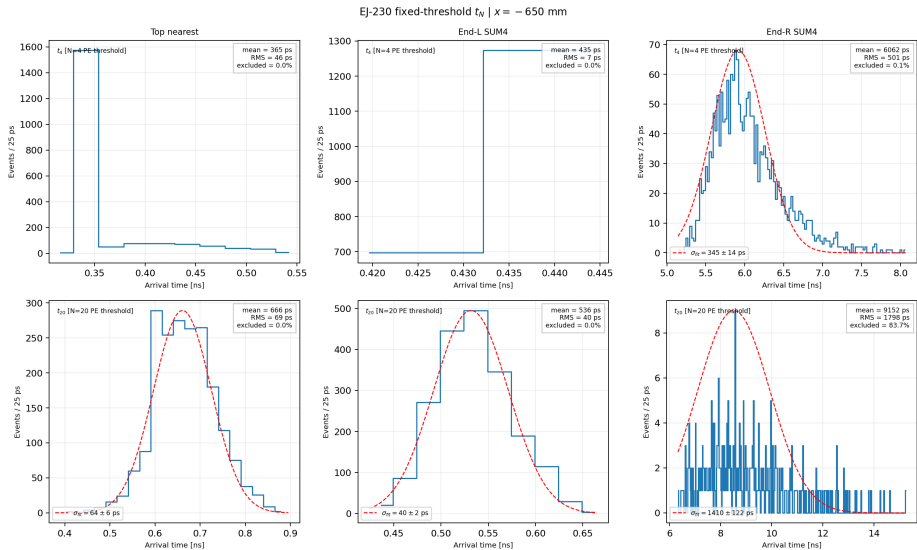
Quantity	Value
End L N_{pe}	2124.3 ± 11.6
End R N_{pe}	29.9 ± 0.2
Nearest Top ID	18 (window-track exception)
Nearest Top N_{pe}	318.0 ± 1.8
Nearest Top $\langle t \rangle$	2.3394 ± 0.0020 ns
Nearest Top FPT	0.34248 ± 0.00020 ns

Run 2 | $x = -650$ mm | Photon arrival $\langle N(t) \rangle$

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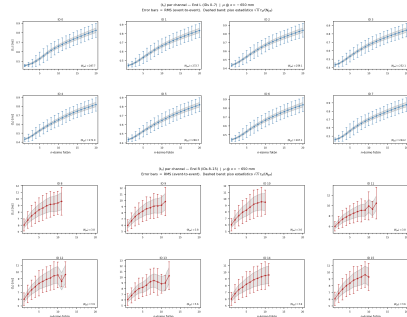
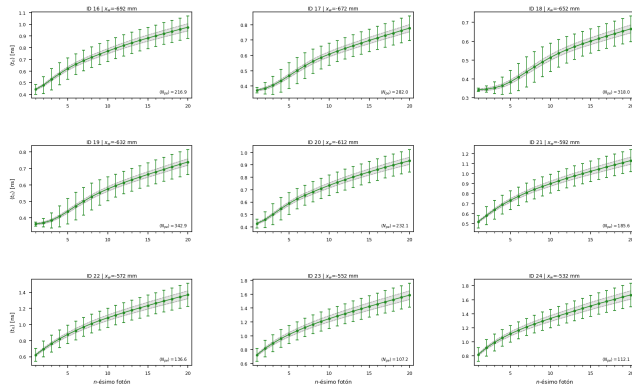


Run 2 | $x = -650$ mm | Time-to-threshold t_N



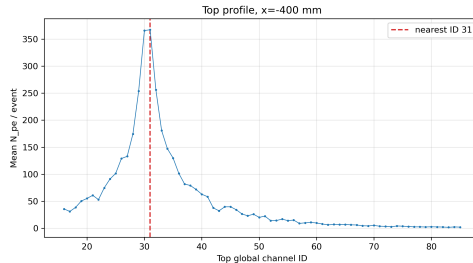
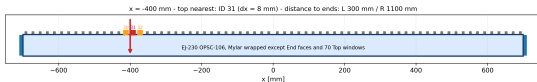
Run 2 | $x = -650$ mm | $\langle t_n \rangle$ per channel (dispersion)

$\langle t_n \rangle$ per Top channel (nearest ± 4) | $\mu @ x = -650$ mm
Error bars = RMS. Dashed band: piso estadístico $\sqrt{n} \tau_d / \langle N_{pe} \rangle$



Error bars = RMS (event-to-event). Dashed band: statistical floor $\sigma_{stat}(t_n) \approx \sqrt{n} \tau_d / \langle N_{pe} \rangle$ ($\tau_d = 1.5$ ns). See dispersion-decomposition frame for physical interpretation.

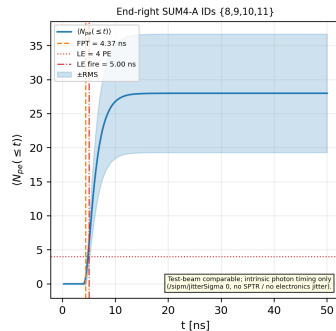
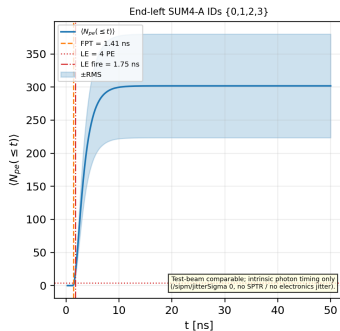
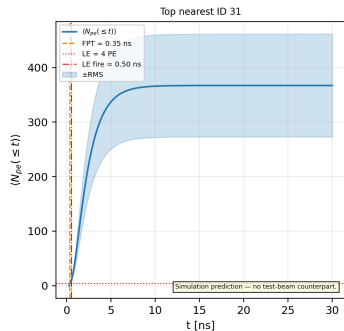
Position $x=-400$ mm



Quantity	Value
End L N_{pe}	581.1 ± 3.4
End R N_{pe}	53.0 ± 0.3
Nearest Top ID	31 (exact geometry)
Nearest Top N_{pe}	367.2 ± 2.1
Nearest Top $\langle t \rangle$	2.4926 ± 0.0020 ns
Nearest Top FPT	0.34578 ± 0.00019 ns

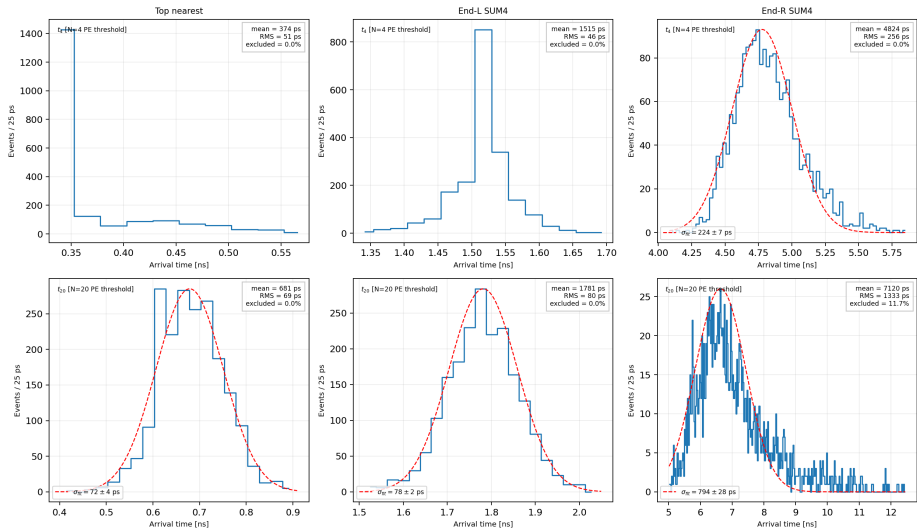
Run 3 | $x = -400$ mm | Photon arrival $\langle N(t) \rangle$

$x = -400$ mm | Photon arrival $\langle N(t) \rangle$



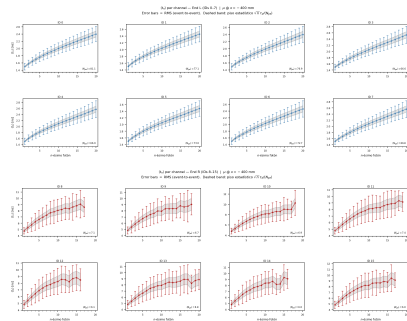
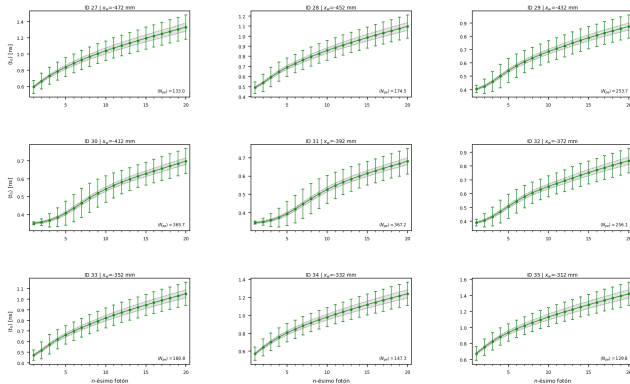
Run 3 | $x = -400$ mm | Time-to-threshold t_N

EJ-230 fixed-threshold t_N | $x = -400$ mm



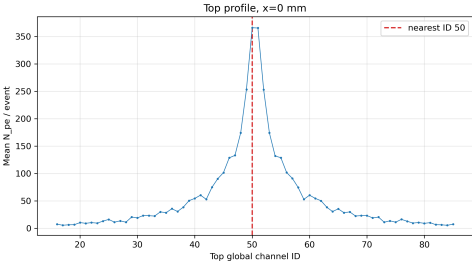
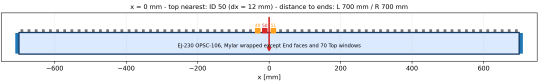
Run 3 | $x = -400$ mm | $\langle t_n \rangle$ per channel (dispersion)

$\langle t_n \rangle$ per Top channel (nearest ± 4) | $\mu @ x = -400$ mm
Error bars = RMS. Dashed band: piso estadístico $\sqrt{n} \tau_d / \langle N_{pe} \rangle$



Error bars = RMS (event-to-event). Dashed band: statistical floor $\sigma_{stat}(t_n) \approx \sqrt{n} \tau_d / \langle N_{pe} \rangle$ ($\tau_d = 1.5$ ns). See dispersion-decomposition frame for physical interpretation.

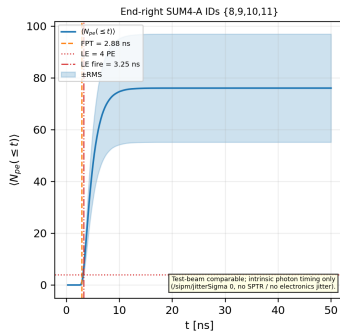
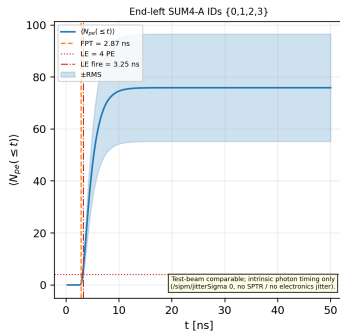
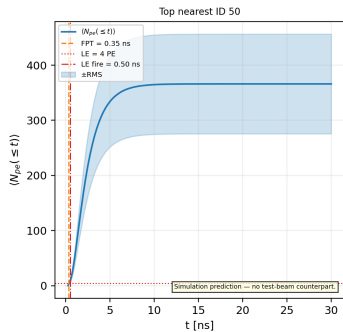
Position $x=0$ mm



Quantity	Value
End L N_{pe}	151.0 ± 0.9
End R N_{pe}	151.1 ± 0.9
Nearest Top ID	50 (exact geometry)
Nearest Top N_{pe}	366.0 ± 2.0
Nearest Top $\langle t \rangle$	2.5528 ± 0.0020 ns
Nearest Top FPT	0.35099 ± 0.00023 ns

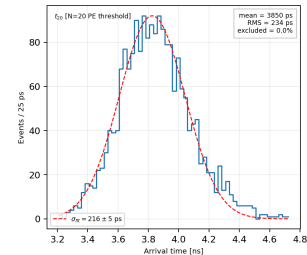
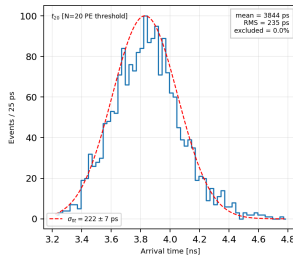
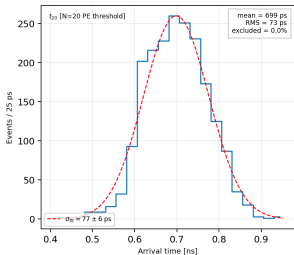
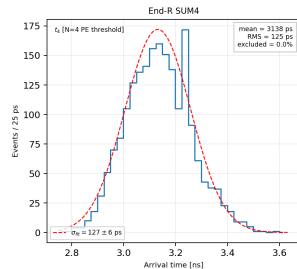
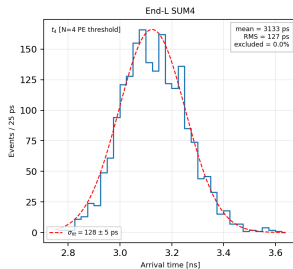
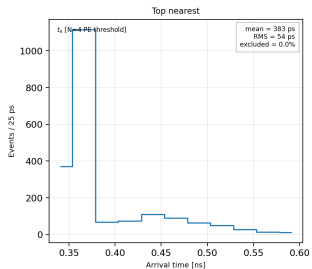
Run 4 | $x = +0$ mm | Photon arrival $\langle N(t) \rangle$

$x = +0$ mm | Photon arrival $\langle N(t) \rangle$



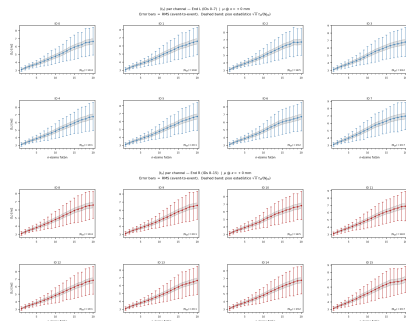
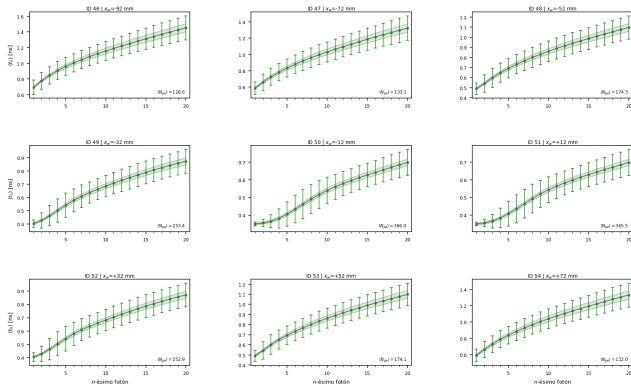
Run 4 | $x = +0$ mm | Time-to-threshold t_N

EJ-230 fixed-threshold t_W | $x = +0$ mm



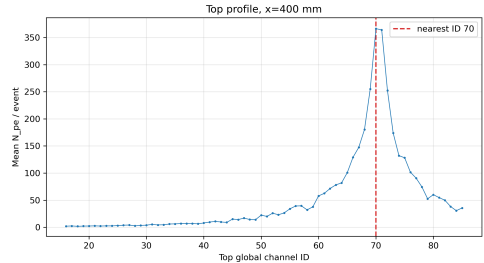
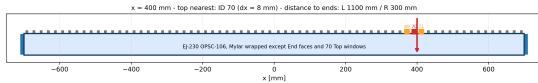
Run 4 | $x = +0$ mm | $\langle t_n \rangle$ per channel (dispersion)

t_d per Top channel (nearest ± 4) | $\mu @ x = +0$ mm
Error bars = RMS. Dashed band: piso estadístico $\sqrt{n} \tau_d / \langle N_{pe} \rangle$



Error bars = RMS (event-to-event). Dashed band: statistical floor $\sigma_{stat}(t_n) \approx \sqrt{n} \tau_d / \langle N_{pe} \rangle$ ($\tau_d = 1.5$ ns). See dispersion-decomposition frame for physical interpretation.

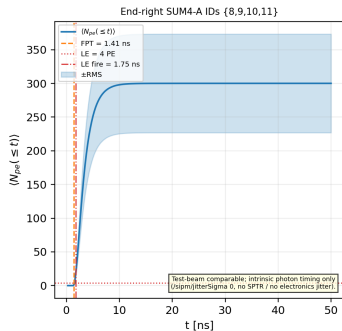
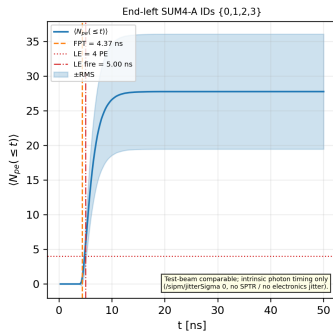
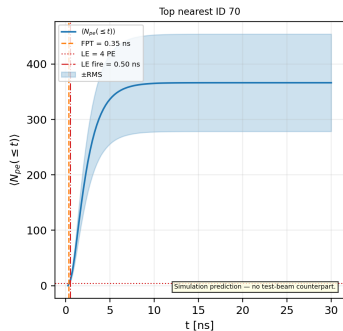
Position $x=400$ mm



Quantity	Value
End L N_{pe}	52.8 ± 0.3
End R N_{pe}	577.8 ± 3.1
Nearest Top ID	70 (exact geometry)
Nearest Top N_{pe}	366.3 ± 2.0
Nearest Top $\langle t \rangle$	2.4909 ± 0.0020 ns
Nearest Top FPT	0.34615 ± 0.00025 ns

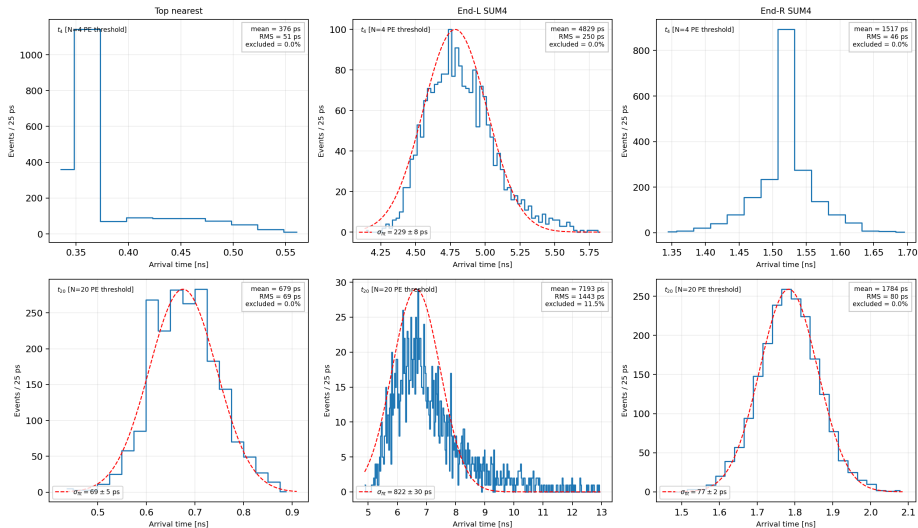
Run 5 | $x = +400$ mm | Photon arrival $\langle N(t) \rangle$

$x = +400$ mm | Photon arrival $\langle N(t) \rangle$

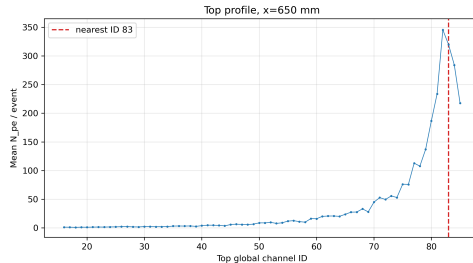
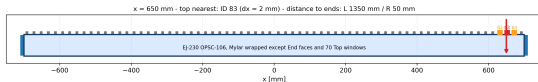


Run 5 | $x = +400$ mm | Time-to-threshold t_N

EJ-230 fixed-threshold t_N | $x = +400$ mm



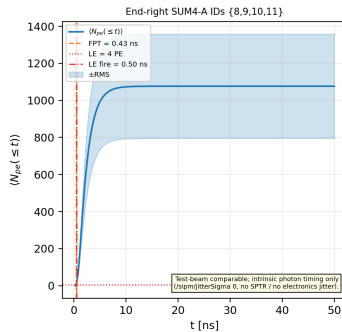
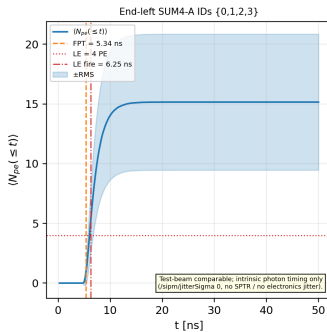
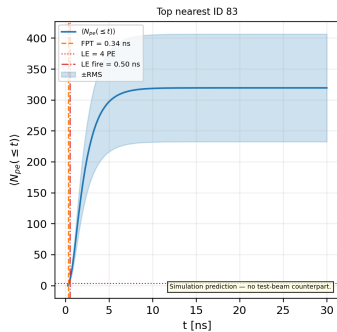
Position $x=650$ mm



Quantity	Value
End L N_{pe}	30.2 ± 0.2
End R N_{pe}	2133.3 ± 12.4
Nearest Top ID	83 (window-track exception)
Nearest Top N_{pe}	319.7 ± 1.9
Nearest Top $\langle t \rangle$	2.3425 ± 0.0020 ns
Nearest Top FPT	0.34249 ± 0.00015 ns

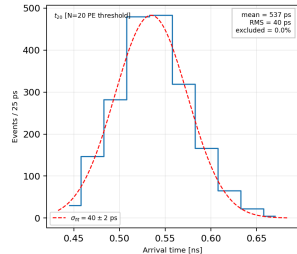
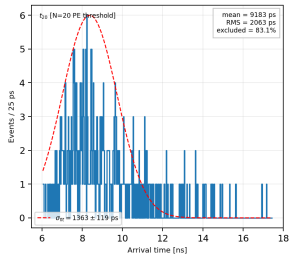
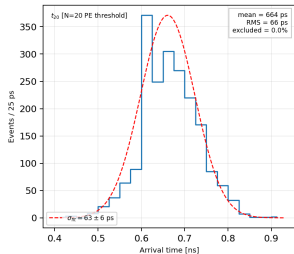
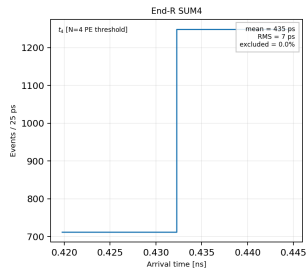
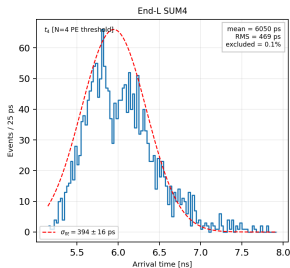
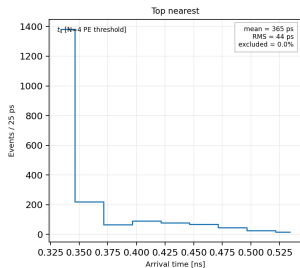
Run 6 | $x = +650$ mm | Photon arrival $\langle N(t) \rangle$

$x = +650$ mm | Photon arrival $\langle N(t) \rangle$



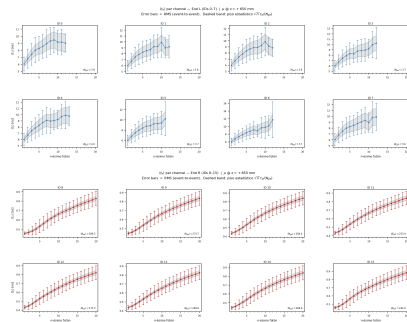
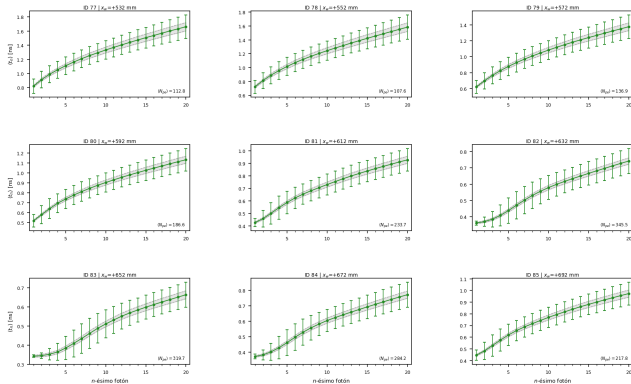
Run 6 | $x = +650$ mm | Time-to-threshold t_N

EJ-230 fixed-threshold t_N | $x = +650$ mm



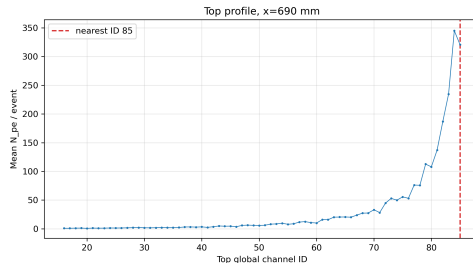
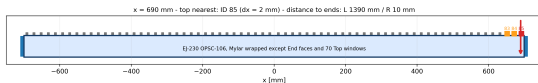
Run 6 | $x = +650$ mm | $\langle t_n \rangle$ per channel (dispersion)

$\langle t_n \rangle$ per Top channel (nearest ± 4) | $\mu @ x = +650$ mm
Error bars = RMS. Dashed band: piso estadístico $\sqrt{n} \tau_d / \langle N_{pe} \rangle$



Error bars = RMS (event-to-event). Dashed band: statistical floor $\sigma_{stat}(t_n) \approx \sqrt{n} \tau_d / \langle N_{pe} \rangle$ ($\tau_d = 1.5$ ns). See dispersion-decomposition frame for physical interpretation.

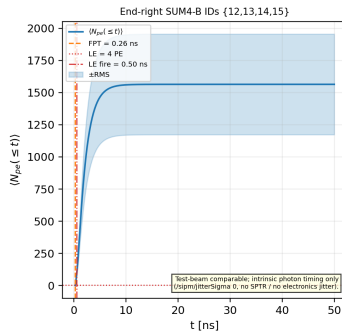
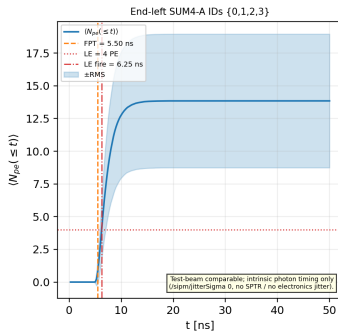
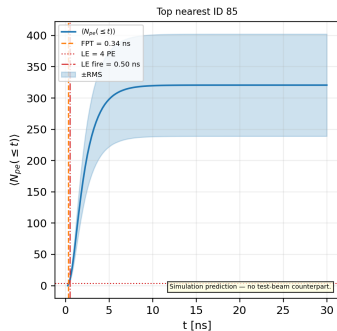
Position $x=690$ mm



Quantity	Value
End L N_{pe}	27.6 ± 0.2
End R N_{pe}	3089.6 ± 17.1
Nearest Top ID	85 (window-track exception)
Nearest Top N_{pe}	320.6 ± 1.8
Nearest Top $\langle t \rangle$	2.3455 ± 0.0020 ns
Nearest Top FPT	0.34224 ± 0.00015 ns

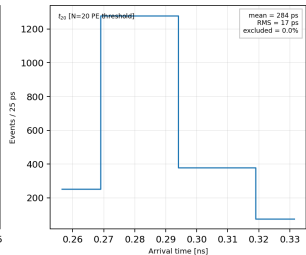
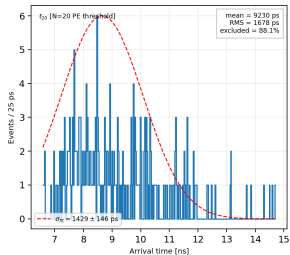
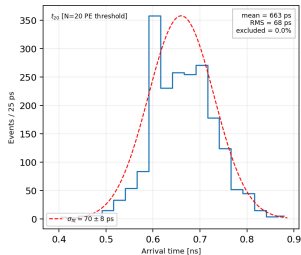
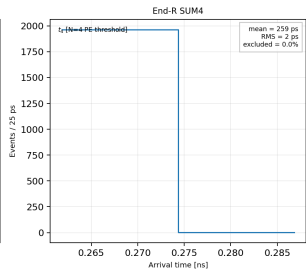
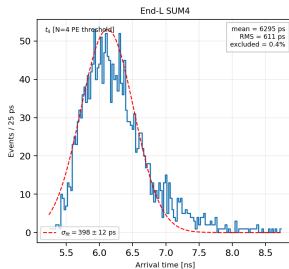
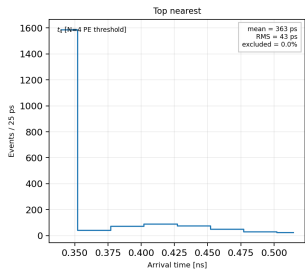
Run 7 | $x = +690$ mm | Photon arrival $\langle N(t) \rangle$

$x = +690$ mm | Photon arrival $\langle N(t) \rangle$



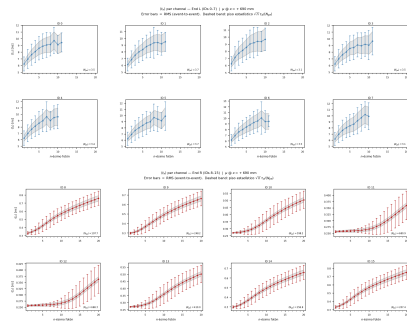
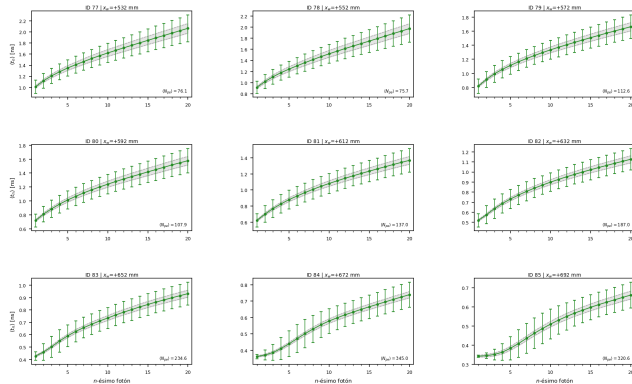
Run 7 | $x = +690$ mm | Time-to-threshold t_N

EJ-230 fixed-threshold t_N | $x = +690$ mm



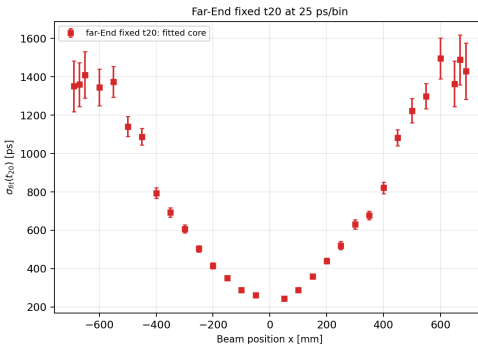
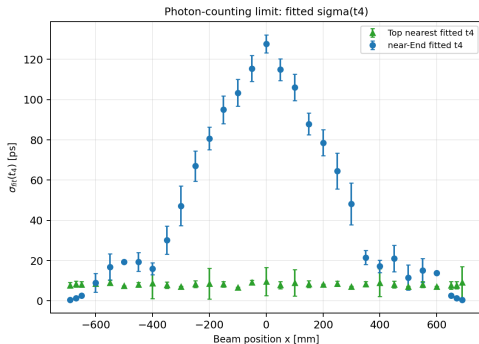
Run 7 | $x = +690$ mm | $\langle t_n \rangle$ per channel (dispersion)

$\langle t_n \rangle$ per Top channel (nearest ± 4) | $\mu @ x = +690$ mm
Error bars = RMS. Dashed band: piso estadístico $\sqrt{n} \tau_d / \langle N_{pe} \rangle$



$\sigma(t_N)$ vs beam position: synthesis

EJ-230 fixed-threshold timing synthesis



How to read this plot

Why: Separate the fitted photon-counting core $\sigma_{fit}(t_4)$ from the fixed-threshold far-End t_{20} display.

Axes: x is beam position; y is the fitted Gaussian-core width; all key-position fixed t_{20} cores have valid Gaussian fits.

Takeaway: Across key positions, nearest-Top fitted $\sigma(t_4)$ spans 7.8–9.6 ps and near-End fitted $\sigma(t_4)$ spans 0.6–127.7 ps (127.7 ps at center). Fixed t_{20} is fitable at 6/6 key far-End positions; unfitable cores retain their measured histogram and RMS, with adaptive N_{eff} in Backup.

Why does t_n disperse? Decomposition into four mechanisms

1. **Order statistics on the emission tail (irreducible):** t_n are order statistics of N_{pe} draws from the emission pdf ($\tau_r = 0.5$ ns, $\tau_d = 1.5$ ns); successive gaps $\sim \text{Exp}(\tau_d/(N_{pe} - n + 1))$, $\sigma_{\text{stat}}(t_n) \approx \sqrt{n} \tau_d / N_{pe}$ for $n \ll N_{pe}$ — grows monotonically with n even with perfect optics.
2. **Landau N_{pe} fluctuations:**
 $\text{Var}_L(t_n) = (\partial \langle t_n \rangle / \partial N_{pe})^2 \text{Var}(N_{pe})$ with $\text{Var}(N_{pe})$ from the documented $F = 1 + c \langle N_{pe} \rangle$. Statistical but non-Poissonian.
3. **Optical path-length dispersion:** TIR photons at $n = 1.58$ ($\theta_c \approx 39^\circ$) span axial speeds $c/(n \cos \theta) \rightarrow 1.4\text{--}1.5$ ns/m per metre plus Mylar bounce tail ($R = 0.98$). Low- n photons: quasi-direct; high- n : reflection-dominated $\Rightarrow \text{RMS} \propto n$.
4. **Attenuation truncation:** distant channels ($\lambda_{\text{eff}} \approx 30.8$ cm) populate only $n = 1\text{--}3$ with large purely statistical bars.

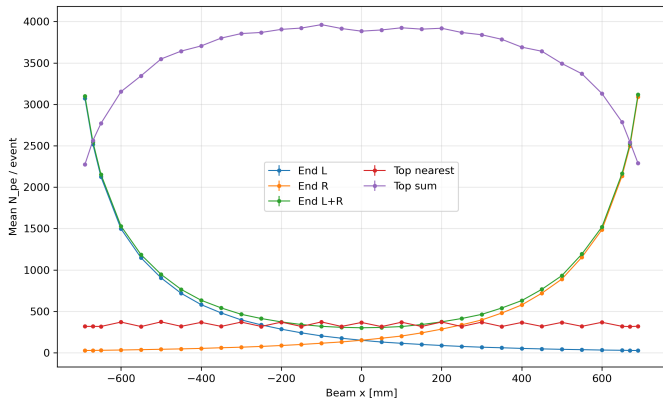
Takeaway

$\text{RMS}(t_n) - \sqrt{n} \tau_d / N_{pe}$ isolates the optical-propagation contribution — the per-channel diagnostic in the $\langle t_n \rangle$ dispersion frames and the same dispersion that limits σ_{group} .

Simulation only / intrinsic

Top: simulation prediction, no test-beam counterpart. End: test-beam comparable, intrinsic precision (no SPTR / no jitter).

Photon budget versus beam position



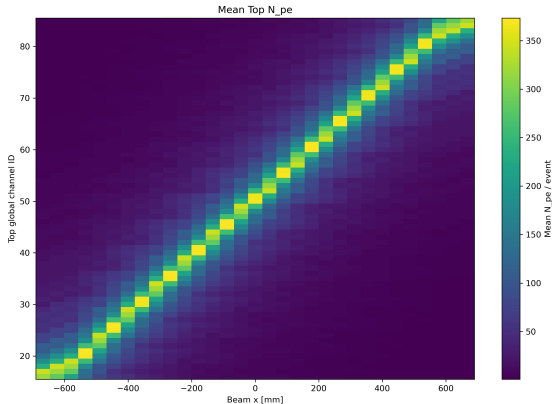
How to read this plot

Why: Quantify N_{pe} collected at each End and nearest-Top channel as the muon track scans the bar.

Axes: x: beam position [mm]; y: mean N_{pe} per event (End sums left/right and nearest Top).

Takeaway: N_{pe} decreases exponentially with track-to-End distance ($\lambda_{eff} \approx 30.8$ cm $\ll$ ABSLENGTH = 120 cm, system effective length); nearest-Top N_{pe} peaks directly under the SiPM window.

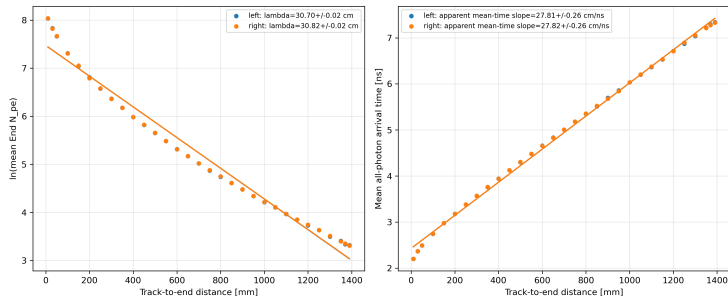
Top localization: full coverage diagonal



CSV-derived localization gate

31/31 positions pass. 18 positions use the documented window-track rule; the maximum nearest-channel deficit is 8.37%. All remaining positions have their strict maximum at the geometrically nearest channel.

Attenuation and the historical mean-time slope



$\lambda_{\text{eff},L} = 30.70 \pm 0.02 \text{ cm}$, $\lambda_{\text{eff},R} = 30.82 \pm 0.02 \text{ cm}$; the mean-time slope is tail-biased and is not a propagation velocity.

How to read this plot

Why: Measure the effective attenuation length λ_{eff} and test whether the mean-time slope encodes a light-propagation velocity.

Axes: Left: mean N_{pe} vs track-to-End distance [cm]; exponential fit gives λ_{eff} . Right: mean arrival time [ns] vs same distance.

Takeaway: $\lambda_{\text{eff}} \approx 30.8 \text{ cm} \ll \text{ABSLENGTH} = 120 \text{ cm}$: λ_{eff} is a system effective length combining bulk absorption, Mylar reflection losses, and Top window drainage—not the material property. The mean-time slope is tail-biased; no propagation velocity is reliably extracted.

Why the Fano factor grows with $\langle N_{pe} \rangle$

1. **Pure Poisson (fixed ΔE):** $\text{Var}(N_{pe}) = \langle N_{pe} \rangle$, hence $F \equiv 1$.
2. **Landau fluctuations in ΔE** (law of total variance):

$$\text{Var}(N_{pe}) = \langle N_{pe} \rangle + \left(\frac{\langle N_{pe} \rangle}{\langle \Delta E \rangle} \right)^2 \text{Var}(\Delta E),$$

$\Rightarrow F = 1 + c \langle N_{pe} \rangle$, $c = \text{Var}(\Delta E) / \langle \Delta E \rangle^2$
(collection-efficiency terms absorbed in c).

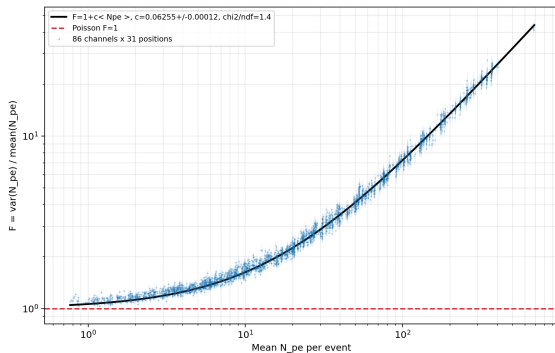
3. **Poisson limit $F \rightarrow 1$:** only at $\langle N_{pe} \rangle \rightarrow 0$
(counting noise dominates over energy-loss fluctuations).
4. **Effective relative fluctuation:** $\sqrt{c} \approx 0.250$ is the combined energy-loss + collection jitter, consistent with a thin (10 mm) plastic absorber.

Fit result (see next slide)

$$c = 0.062547 \pm 0.000117$$

The *same* Landau ΔE fluctuation that shapes the asymmetric N_{pe} spectrum (next section) also generates the super-Poissonian Fano factor: **one physical mechanism, two observables.**

Landau dominance: Fano factor versus mean N_{pe}



$F = 1 + c \langle N_{pe} \rangle$, $c = 0.062547 \pm 0.000117$, $\chi^2/ndf = 1.41$. Poisson $F = 1$ is approached only at low light.

How to read this plot

Why: Measure $F = \text{Var}(N_{pe}) / \langle N_{pe} \rangle$ to distinguish Poisson ($F = 1$) from Landau-dominated fluctuations.

Axes: x: $\langle N_{pe} \rangle$ per event; y: $\text{Var} / \langle N_{pe} \rangle$. Dashed: Poisson $F = 1$. Line: linear fit $F = 1 + c \langle N_{pe} \rangle$.

Takeaway: F grows linearly: Landau ΔE fluctuations dominate counting noise at all light levels. $\sqrt{c}$ is the effective relative energy-loss fluctuation of this 10 mm absorber.

Why Landau (and why Moyal as analytic proxy)

Why not Gaussian:

- ▶ In 10 mm of plastic the muon undergoes many collisions, but the energy-transfer cross-section ($d\sigma/d\varepsilon \propto 1/\varepsilon^2$, Rutherford/Møller) has a hard tail.
- ▶ Rare large transfers (δ -rays) dominate the variance; the variance per collision diverges and the Central Limit Theorem does not apply.
- ▶ Result: asymmetric distribution with a hard right tail — the Landau distribution.
- ▶ Gaussian (Vavilov→Gauss) limit only applies to *thick* absorbers.

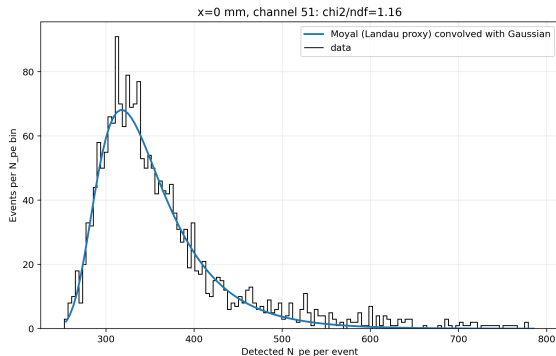
Why Moyal:

- ▶ Landau density has no closed form (contour integral).
- ▶ The Moyal function is a closed-form analytic approximation to the Landau core and early tail, enabling stable MINUIT fits.
- ▶ *Known limitation*: extreme Moyal tail undershoots true Landau tail; fit quality degrades at high N_{pe} .

Moyal $\otimes$ Gauss:

- ▶ Photo-electron counting (Poisson) and collection fluctuations broaden the peak symmetrically.
- ▶ Full model: Moyal $\otimes$ Gaussian.
- ▶ Connects to Fano slide: same Landau ΔE fluctuation, two observables (spectrum shape + super-Poissonian F).

High-light example: Moyal–Gaussian Landau proxy



Moyal approximates Landau. Some channels have poor χ^2 or an unidentifiable Gaussian width; all recorded in `exec10_landau_mpv.csv`.

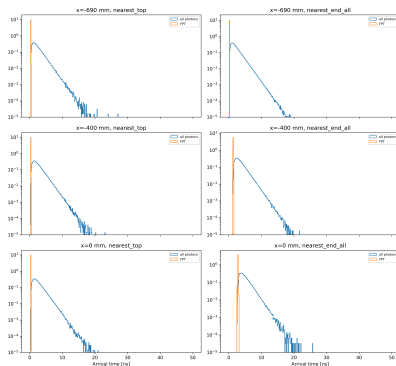
How to read this plot

Why: Fit the N_{pe} spectrum at a high-yield position to quantify the Landau MPV and width.

Axes: x: N_{pe} per event; y: counts. Data (step), Moyal $\otimes$ Gauss fit (curve).

Takeaway: Core described by Moyal proxy; hard right tail from δ -rays. MPVs are future inputs for test-beam ToT calibration.

Arrival-time examples: absolute rate and normalized shape



Left: entries per event per ns. Right: area-normalized distributions. FPT = first detected photon time per event.

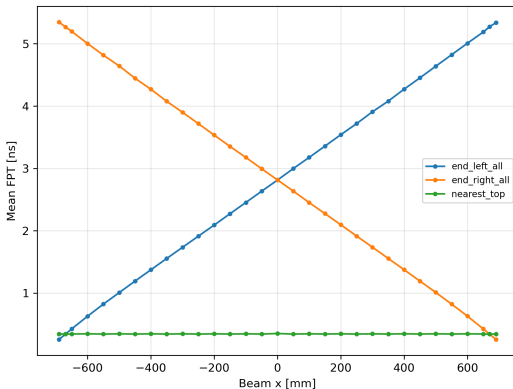
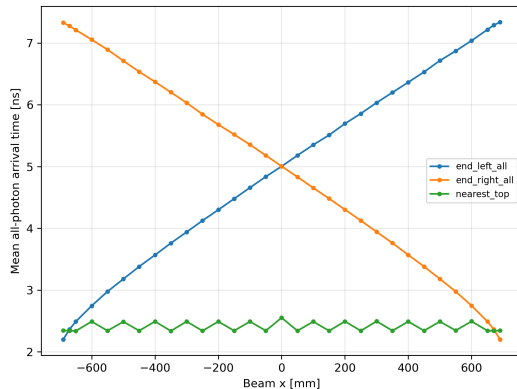
How to read this plot

Why: Illustrate the arrival-time structure of individual Top channels and contrast absolute rate vs shape.

Axes: x: arrival time [ns]; y: left is entries/event/ns, right is area-normalized density [1/ns].

Takeaway: FPT distribution captures the prompt scintillation leading edge; the broad tail reflects multiple-reflection photons.

Mean arrival time and FPT versus x



How to read this plot

Why: Track how the mean photon arrival time and FPT scale with beam position.

Axes: x: beam position [mm]; y: mean arrival time [ns] and mean FPT [ns].

Takeaway: Both estimators increase with track-to-End distance; the mean is more tail-biased than FPT and yields a larger apparent slope.

Nearest-Top Landau/Moyal MPVs at key positions

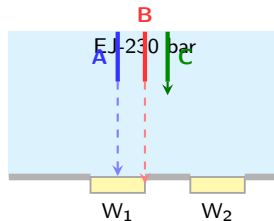
x [mm]	channel	MPV [Npe]	Landau width [Npe]	χ^2/ndf
-690	16	275.0 ± 0.9	23.3 ± 0.8	1.61
-650	18	275.6 ± 0.9	23.7 ± 0.8	1.58
-400	31	317.5 ± 1.0	26.8 ± 0.9	1.56
+0	50	318.5 ± 1.0	26.0 ± 0.9	1.17
+400	70	320.1 ± 1.0	25.3 ± 0.8	1.23
+650	83	275.0 ± 0.9	24.7 ± 0.8	1.87
+690	85	278.1 ± 0.9	24.6 ± 0.8	2.15

Interpretation

The right tail originates primarily from Landau-like energy-loss fluctuations; Poisson counting dominates only in the few-PE limit.

Window-track collection effect: mechanism

- ▶ Top SiPMs couple through index-matched windows ($6 \times 6 \text{ mm}^2$) in the +Y Mylar face.
- ▶ First-pass photon collection depends on the solid angle subtended by the window from the muon track (geometric working hypothesis, consistent with five-run pattern).
- ▶ Empirically, a track at $\pm 2 \text{ mm}$ from the window centre changes the local contrast by 7.9% with minimum significance 9.3σ . The measured centered and midpoint contrast significances are -1.02σ and -0.06σ , respectively.
- ▶ Scan positions with $x \equiv 10 \pmod{20} \text{ mm}$ fall at $\pm 2 \text{ mm}$ from a window centre — hence the “(window-track exception)” label in the tables.
- ▶ Effect is local, symmetric, characterised and bounded: localization diagonal and channel map remain valid.

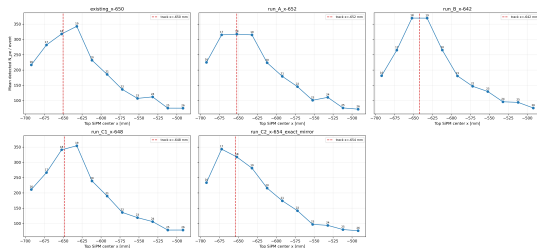


Consequence for the scan

Track B subtends a larger first-pass solid angle to W_1 ; fewer multiple reflections $\Rightarrow$ higher N_{pe} at the nearest channel. At midpoint (C) both windows contribute equally and the effect is null.

Window-track collection effect: five-run test

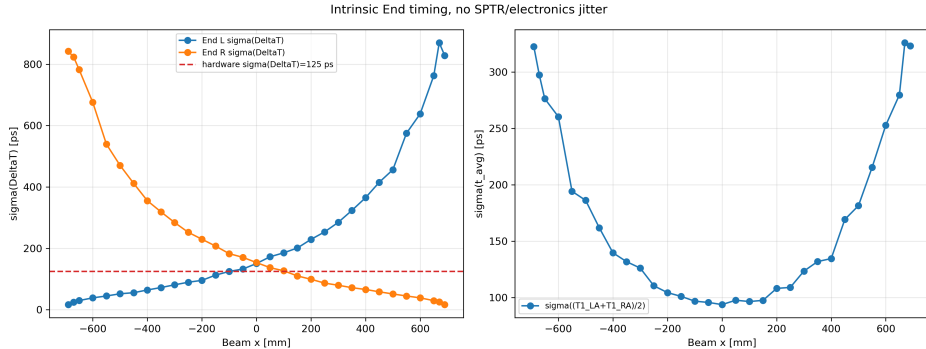
Run	contrast [Npe]	significance
Existing -650	24.96 ± 2.57	9.72
A -652 center	-2.44 ± 2.39	-1.02
B -642 midpoint	-0.19 ± 3.04	-0.06
C1 -648 (+4 mm)	13.27 ± 2.73	4.85
C2 -654 mirror	25.54 ± 2.76	9.26



Finding

Local symmetric ± 2 mm effect: 7.9% with significance 9.3–9.7 σ ; centered/midpoint contrast significances: -1.02 σ /-0.06 σ .

Intrinsic End SUM4 timing



How to read this plot

Why: Measure intrinsic Same-End SUM4 coincidence timing without SPTR or electronics jitter.

Axes: x is beam position; y is $\sigma_{group} = \sigma(\Delta T_{AB})/\sqrt{2}$ for the left and right End groups.

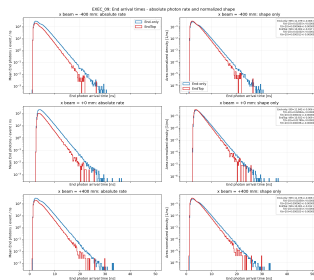
Takeaway: The nearer End improves with its larger photon budget, consistent with $1/\sqrt{N_{pe}}$ scaling. Intrinsic End σ_{group} at $x = 0$: 107.6 ps; full-scan mean 177 ps and range 12–615 ps.

End arrival-time tails: drainage persists, but EndTop widens End timing

x	side	EndTop [ps]	End-only [ps]	ratio
-400	L	45.5 ± 0.7	37.2 ± 0.3	1.223 ± 0.021
-400	R	251.3 ± 4.0	128.0 ± 0.9	1.963 ± 0.034
+0	L	106.5 ± 1.7	76.4 ± 0.5	1.394 ± 0.024
+0	R	108.7 ± 1.7	75.3 ± 0.5	1.443 ± 0.025
+400	L	258.2 ± 4.1	131.5 ± 0.9	1.963 ± 0.034
+400	R	46.7 ± 0.7	37.9 ± 0.3	1.233 ± 0.021

Confirmed mechanism

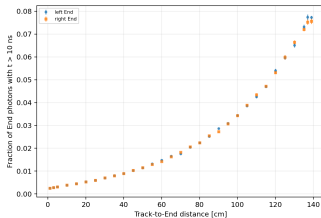
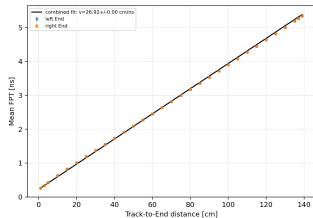
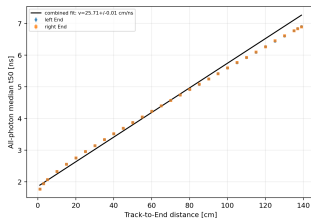
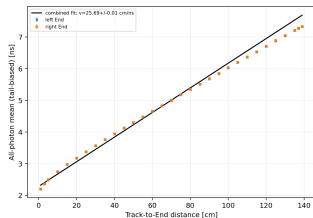
Across $x = 0, \pm 400$ mm, EndTop minus End-only t_{99} ranges from -1.08 to -0.92 ns; the minimum absolute significance among the two late-fraction differences is 6.6σ .



Apples-to-apples diagnosis

PHYSICAL after apples-to-apples audit. Both configurations use the same End SUM4 groups and $\sigma_{group} = \sigma(\Delta T_{AB})/\sqrt{2}$ estimator. After bootstrapping End-only to 2000 events, the ratio spans 1.223 ± 0.019 to 1.963 ± 0.033 ; the smallest 95% lower quantile is 1.186. End-only retains $1.62\text{--}3.04\times$ more End photoelectrons. EndTop still shortens side t_{99} by 0.518–1.182 ns, but the photon loss dominates the timing width for EJ-230.

Apparent timing slopes: no propagation estimator passes



Estimator	slope-derived [cm/ns]	χ^2/ndf
all-photon mean	25.694 ± 0.005	3357.5
all-photon t_{50}	25.714 ± 0.005	4143.2
mean FPT	26.922 ± 0.003	742.6

How to read this plot

Why: Test whether an arrival-time estimator reveals a light-propagation velocity.

Axes: x is track-to-End distance; y is the timing estimator.

Takeaway: Every global line fit is rejected. The reported slopes are apparent velocities dominated by estimator-dependent late-tail bias.

Top readout: analytic timing estimate only

x [mm]	N_{pe}	$\sigma_{t,est}$ [ps]
-690	318.7 ± 1.8	48.51 ± 0.14
-650	318.0 ± 1.8	48.57 ± 0.14
-400	367.2 ± 2.1	45.19 ± 0.13
+0	366.0 ± 2.0	45.27 ± 0.13
+400	366.3 ± 2.0	45.25 ± 0.12
+650	319.7 ± 1.9	48.44 ± 0.15
+690	320.6 ± 1.8	48.36 ± 0.14

Definition, derivation, and scope

$$\sigma_{t,est} = \sqrt{\tau_r \tau_d} / \sqrt{N_{pe}} = 0.866 \text{ ns} / \sqrt{N_{pe}}.$$

Double-exponential emission with $\tau_r = 0.5$ ns and $\tau_d = 1.5$ ns (Knoll, ch. 8). **Analytic orientation only**; the Top readout has no test-beam counterpart in this study.

$\sigma_{t,est}$ vs. simulated $\sigma(t_4)$

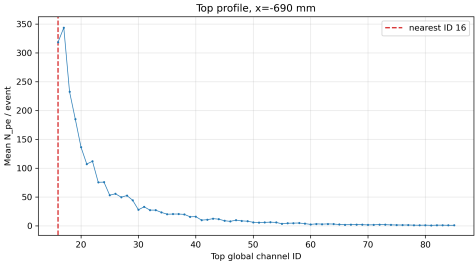
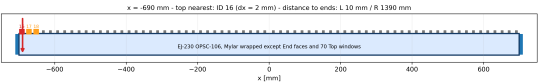
x	$\sigma_{t,est}$ [ps]	fitted $\sigma(t_4)$ [ps]
-690	48.51 ± 0.14	7.8 ± 1.4
-650	48.57 ± 0.14	8.2 ± 1.5
-400	45.19 ± 0.13	8.7 ± 7.5
+0	45.27 ± 0.13	9.6 ± 7.0
+400	45.25 ± 0.12	9.0 ± 6.9
+650	48.44 ± 0.15	7.8 ± 1.9
+690	48.36 ± 0.14	9.2 ± 7.8

Numerical conclusions

EJ-230 synthesis: four findings and numerical checks

- ▶ At center: End sum 302.1 ± 1.2 , nearest Top 366.0 ± 2.0 ; at $x = -690$ mm: End sum 3100.7 ± 16.8 , nearest Top 318.7 ± 1.8 .
- ▶ λ_{eff} : L 30.70 ± 0.02 cm, R 30.82 ± 0.02 cm.
- ▶ Nearest-Top $F = 1 + (0.062547 \pm 0.000117) \langle N_{pe} \rangle$ ($\chi^2/ndf = 1.41$).
- ▶ Intrinsic End σ_{group} at $x = 0$: 107.6 ps; full-scan mean 177 ps and range 12–615 ps.
- ▶ EndTop/End-only timing ratio: 1.223–1.963. Tail drainage persists, but EndTop widens End timing because its End-photon loss dominates for EJ-230.
- ▶ Fitted near-End $\sigma(t_4)$: 0.6–127.7 ps; fitted nearest-Top $\sigma(t_4)$: 7.8–9.6 ps.
- ▶ Four findings: window-track mechanism characterised; tail drainage and its photon-budget tradeoff confirmed; complete channel coverage/localisation validated; adaptive t_N removes unreachable fixed-threshold fits.

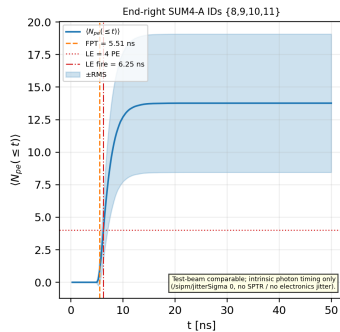
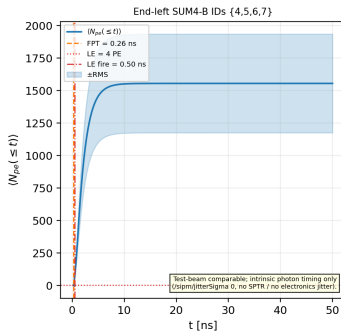
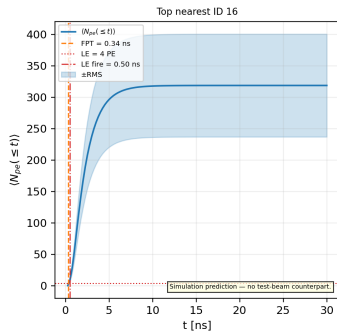
Position $x=-690$ mm



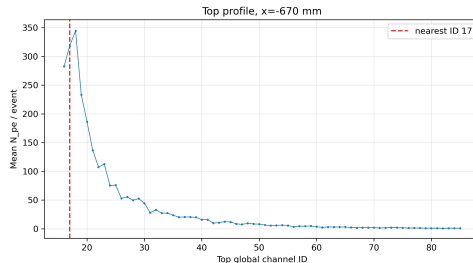
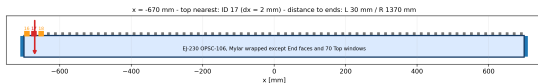
Quantity	Value
End L N_{pe}	3073.3 ± 16.8
End R N_{pe}	27.4 ± 0.2
Nearest Top ID	16 (window-track exception)
Nearest Top N_{pe}	318.7 ± 1.8
Nearest Top $\langle t \rangle$	2.3460 ± 0.0020 ns
Nearest Top FPT	0.34237 ± 0.00012 ns

Run 1 | $x = -690$ mm | Photon arrival $\langle N(t) \rangle$

$x = -690$ mm | Photon arrival $\langle N(t) \rangle$



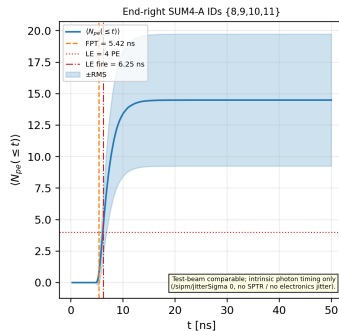
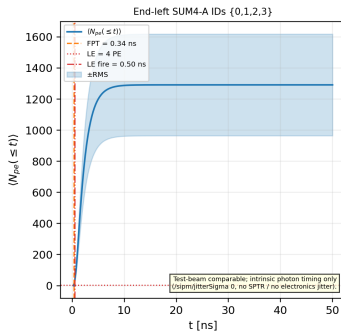
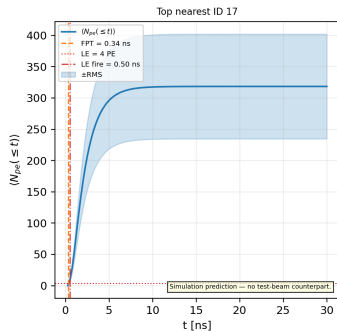
Position $x=-670$ mm



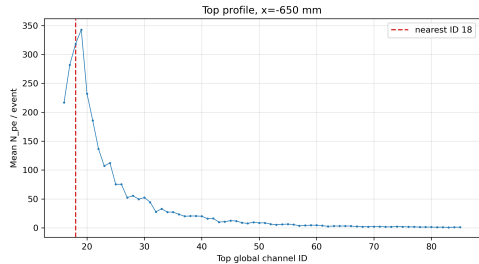
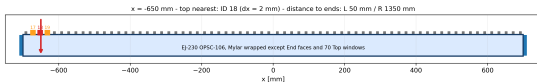
Quantity	Value
End L N_{pe}	2516.7 ± 14.2
End R N_{pe}	28.6 ± 0.2
Nearest Top ID	17 (window-track exception)
Nearest Top N_{pe}	318.5 ± 1.9
Nearest Top $\langle t \rangle$	2.3425 ± 0.0020 ns
Nearest Top FPT	0.34229 ± 0.00020 ns

Run 2 | $x = -670$ mm | Photon arrival $\langle N(t) \rangle$

$x = -670$ mm | Photon arrival $\langle N(t) \rangle$



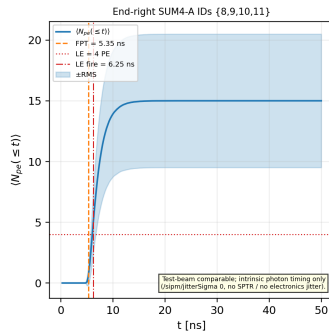
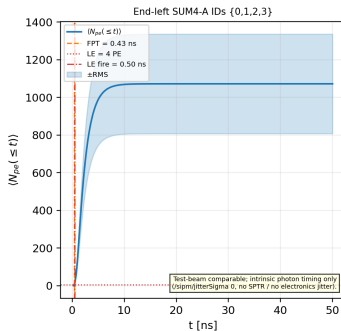
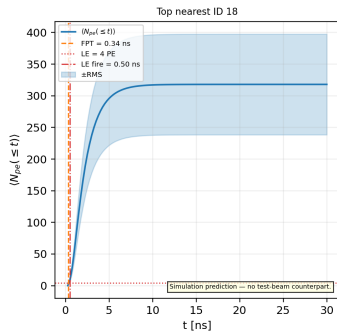
Position $x=-650$ mm



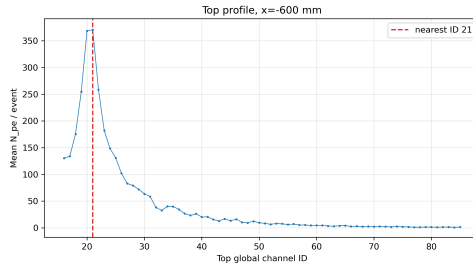
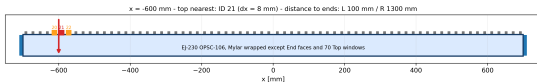
Quantity	Value
End L N_{pe}	2124.3 ± 11.6
End R N_{pe}	29.9 ± 0.2
Nearest Top ID	18 (window-track exception)
Nearest Top N_{pe}	318.0 ± 1.8
Nearest Top $\langle t \rangle$	2.3394 ± 0.0020 ns
Nearest Top FPT	0.34248 ± 0.00020 ns

Run 3 | $x = -650$ mm | Photon arrival $\langle N(t) \rangle$

$x = -650$ mm | Photon arrival $\langle N(t) \rangle$



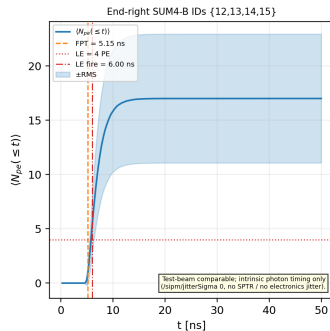
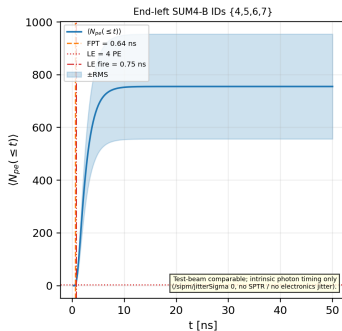
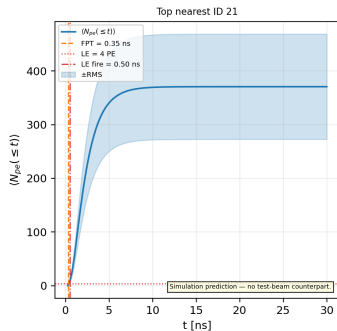
Position $x=-600$ mm



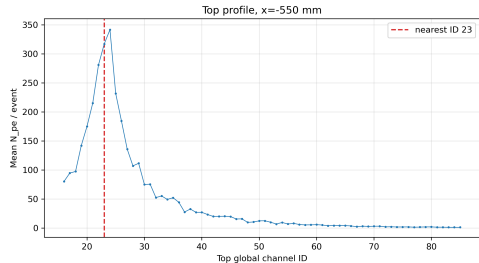
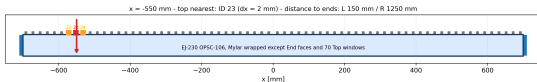
Quantity	Value
End L N_{pe}	1496.7 ± 8.8
End R N_{pe}	33.5 ± 0.2
Nearest Top ID	21 (exact geometry)
Nearest Top N_{pe}	370.7 ± 2.2
Nearest Top $\langle t \rangle$	2.4920 ± 0.0019 ns
Nearest Top FPT	0.34586 ± 0.00023 ns

Run 4 | $x = -600$ mm | Photon arrival $\langle N(t) \rangle$

$x = -600$ mm | Photon arrival $\langle N(t) \rangle$



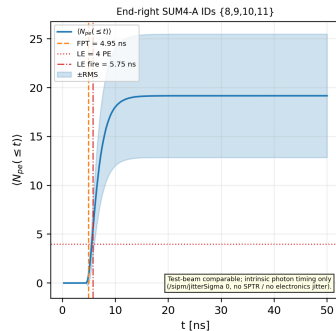
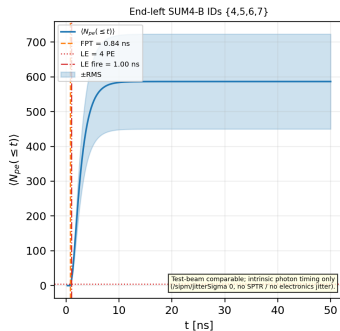
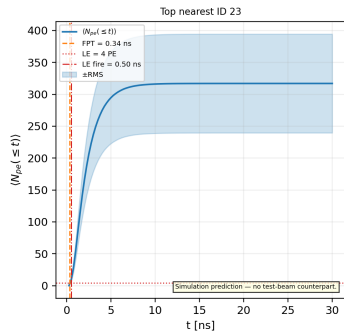
Position $x=-550$ mm



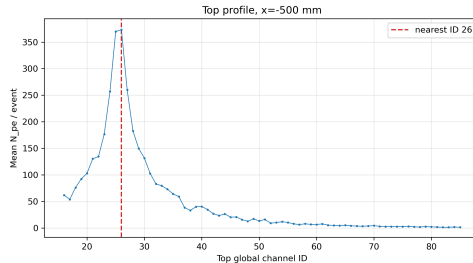
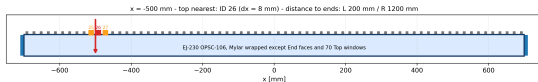
Quantity	Value
End L N_{pe}	1146.8 ± 6.0
End R N_{pe}	37.8 ± 0.2
Nearest Top ID	23 (window-track exception)
Nearest Top N_{pe}	316.9 ± 1.7
Nearest Top $\langle t \rangle$	2.3422 ± 0.0020 ns
Nearest Top FPT	0.34244 ± 0.00014 ns

Run 5 | $x = -550$ mm | Photon arrival $\langle N(t) \rangle$

$x = -550$ mm | Photon arrival $\langle N(t) \rangle$



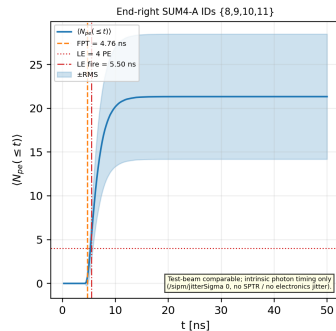
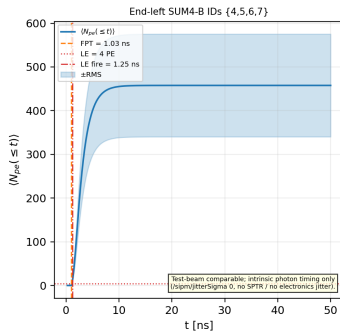
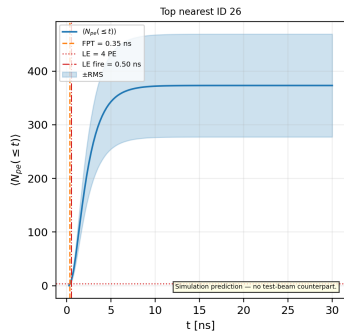
Position $x=-500$ mm



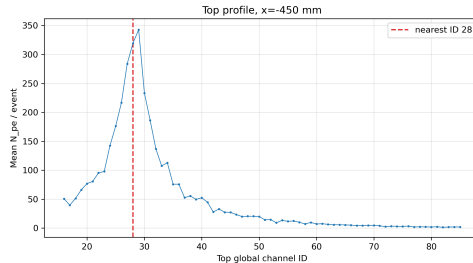
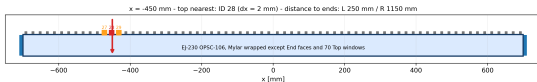
Quantity	Value
End L N_{pe}	904.4 ± 5.2
End R N_{pe}	42.3 ± 0.3
Nearest Top ID	26 (exact geometry)
Nearest Top N_{pe}	373.2 ± 2.1
Nearest Top $\langle t \rangle$	2.4886 ± 0.0019 ns
Nearest Top FPT	0.34538 ± 0.00014 ns

Run 6 | $x = -500$ mm | Photon arrival $\langle N(t) \rangle$

$x = -500$ mm | Photon arrival $\langle N(t) \rangle$



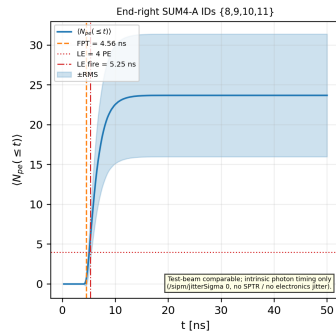
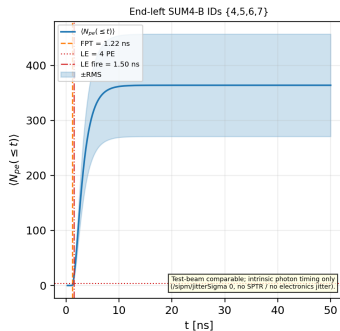
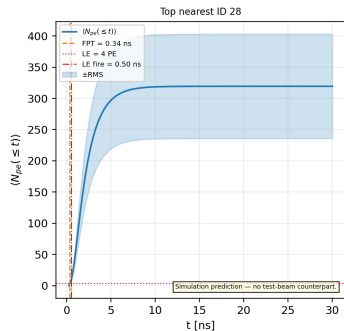
Position $x=-450$ mm



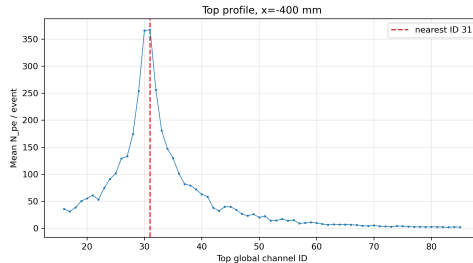
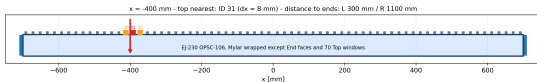
Quantity	Value
End L N_{pe}	718.6 ± 4.1
End R N_{pe}	47.0 ± 0.3
Nearest Top ID	28 (window-track exception)
Nearest Top N_{pe}	319.4 ± 1.9
Nearest Top $\langle t \rangle$	2.3423 ± 0.0020 ns
Nearest Top FPT	0.34239 ± 0.00020 ns

Run 7 | $x = -450$ mm | Photon arrival $\langle N(t) \rangle$

$x = -450$ mm | Photon arrival $\langle N(t) \rangle$



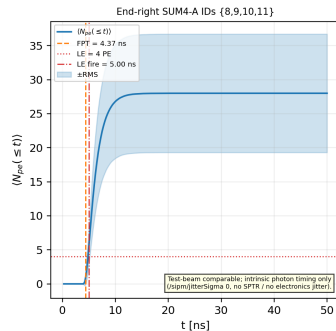
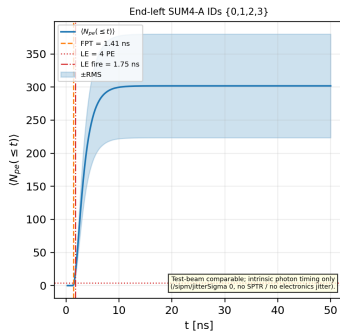
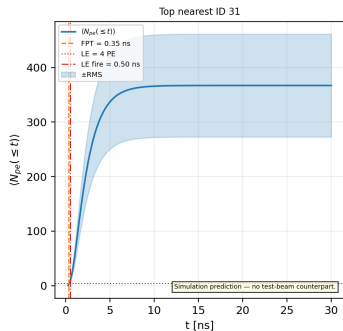
Position $x=-400$ mm



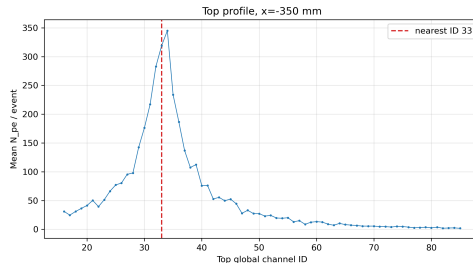
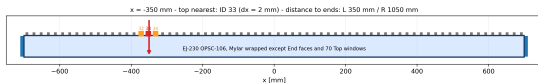
Quantity	Value
End L N_{pe}	581.1 ± 3.4
End R N_{pe}	53.0 ± 0.3
Nearest Top ID	31 (exact geometry)
Nearest Top N_{pe}	367.2 ± 2.1
Nearest Top $\langle t \rangle$	2.4926 ± 0.0020 ns
Nearest Top FPT	0.34578 ± 0.00019 ns

Run 8 | $x = -400$ mm | Photon arrival $\langle N(t) \rangle$

$x = -400$ mm | Photon arrival $\langle N(t) \rangle$



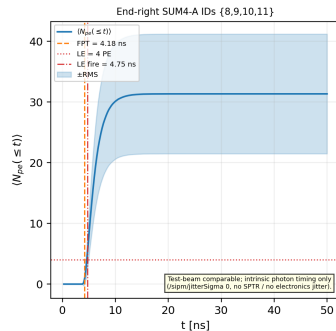
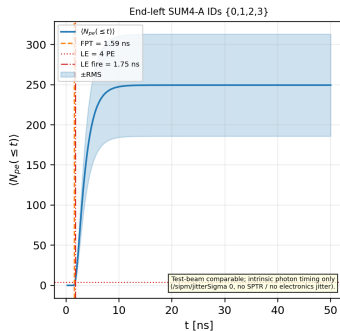
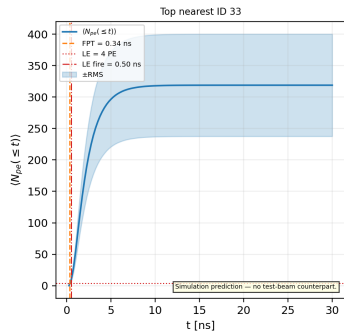
Position $x=-350$ mm



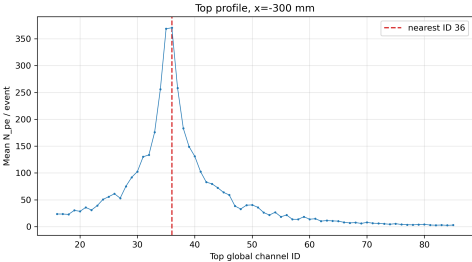
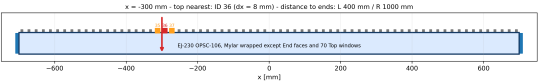
Quantity	Value
End L N_{pe}	481.1 ± 2.7
End R N_{pe}	60.7 ± 0.4
Nearest Top ID	33 (window-track exception)
Nearest Top N_{pe}	318.9 ± 1.8
Nearest Top $\langle t \rangle$	2.3390 ± 0.0020 ns
Nearest Top FPT	0.34245 ± 0.00015 ns

Run 9 | $x = -350$ mm | Photon arrival $\langle N(t) \rangle$

$x = -350$ mm | Photon arrival $\langle N(t) \rangle$



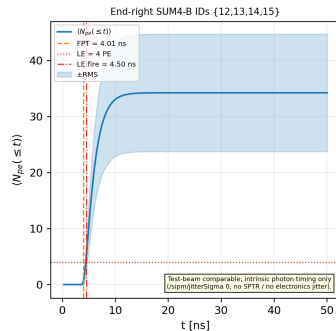
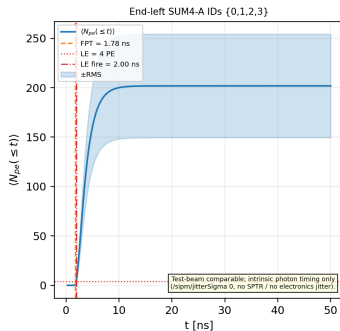
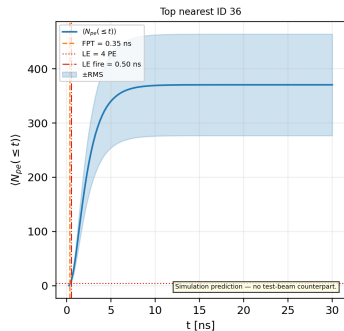
Position $x=-300$ mm



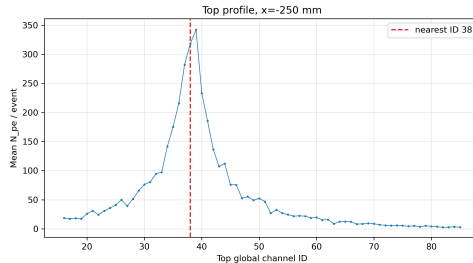
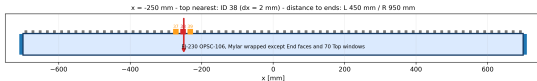
Quantity	Value
End L N_{pe}	397.1 ± 2.3
End R N_{pe}	67.6 ± 0.4
Nearest Top ID	36 (exact geometry)
Nearest Top N_{pe}	370.6 ± 2.1
Nearest Top $\langle t \rangle$	2.4914 ± 0.0019 ns
Nearest Top FPT	0.34579 ± 0.00017 ns

Run 10 | $x = -300$ mm | Photon arrival $\langle N(t) \rangle$

$x = -300$ mm | Photon arrival $\langle N(t) \rangle$



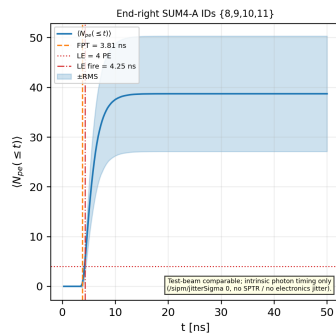
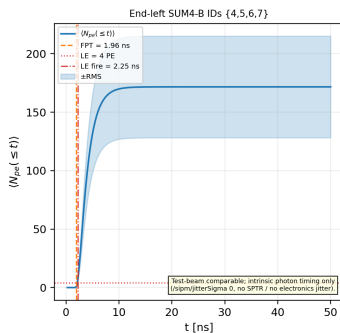
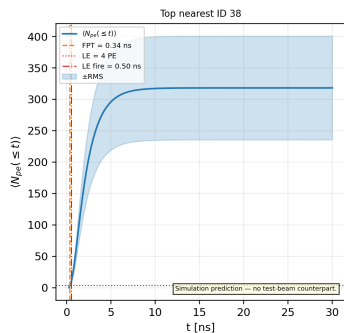
Position $x=-250$ mm



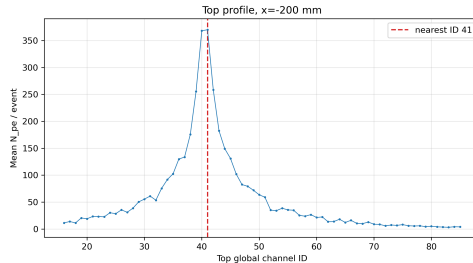
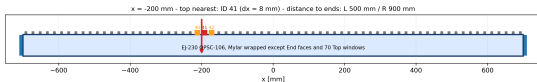
Quantity	Value
End L N_{pe}	336.8 ± 1.9
End R N_{pe}	76.8 ± 0.5
Nearest Top ID	38 (window-track exception)
Nearest Top N_{pe}	318.1 ± 1.8
Nearest Top $\langle t \rangle$	2.3401 ± 0.0020 ns
Nearest Top FPT	0.34234 ± 0.00017 ns

Run 11 | $x = -250$ mm | Photon arrival $\langle N(t) \rangle$

$x = -250$ mm | Photon arrival $\langle N(t) \rangle$



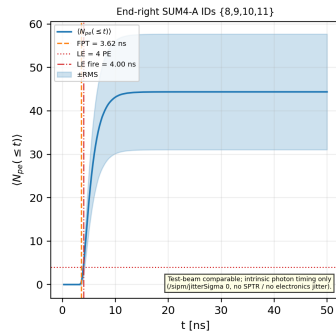
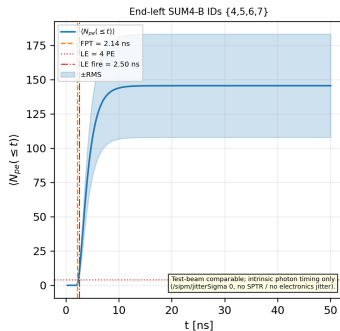
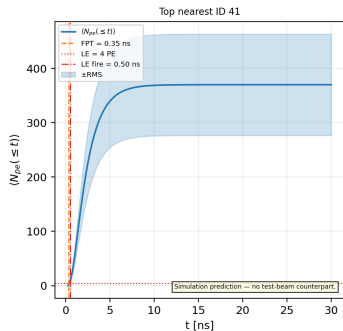
Position $x=-200$ mm



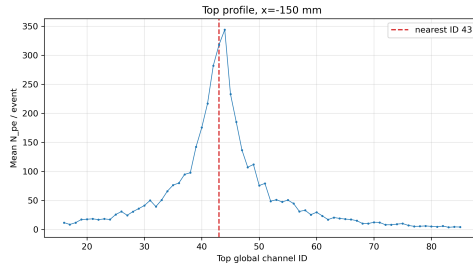
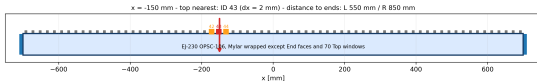
Quantity	Value
End L N_{pe}	285.2 ± 1.6
End R N_{pe}	87.9 ± 0.5
Nearest Top ID	41 (exact geometry)
Nearest Top N_{pe}	370.1 ± 2.1
Nearest Top $\langle t \rangle$	2.4913 ± 0.0019 ns
Nearest Top FPT	0.34568 ± 0.00014 ns

Run 12 | $x = -200$ mm | Photon arrival $\langle N(t) \rangle$

$x = -200$ mm | Photon arrival $\langle N(t) \rangle$



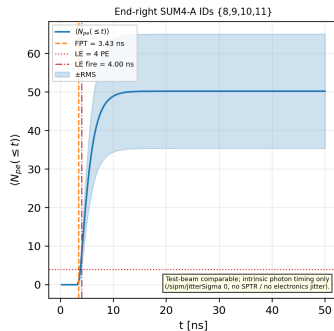
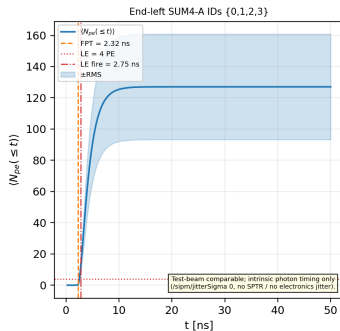
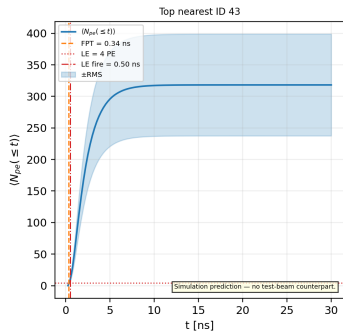
Position $x=-150$ mm



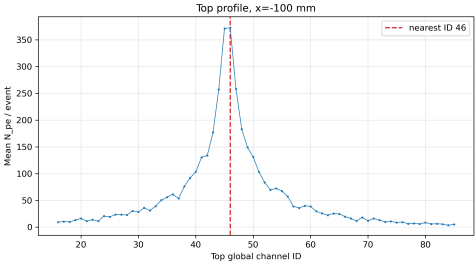
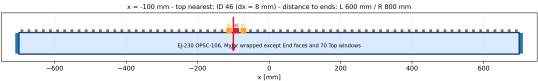
Quantity	Value
End L N_{pe}	241.0 ± 1.4
End R N_{pe}	100.4 ± 0.6
Nearest Top ID	43 (window-track exception)
Nearest Top N_{pe}	318.2 ± 1.8
Nearest Top $\langle t \rangle$	2.3419 ± 0.0020 ns
Nearest Top FPT	0.34245 ± 0.00019 ns

Run 13 | $x = -150$ mm | Photon arrival $\langle N(t) \rangle$

$x = -150$ mm | Photon arrival $\langle N(t) \rangle$



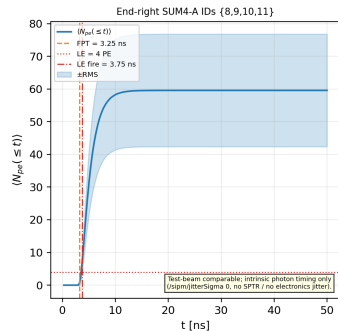
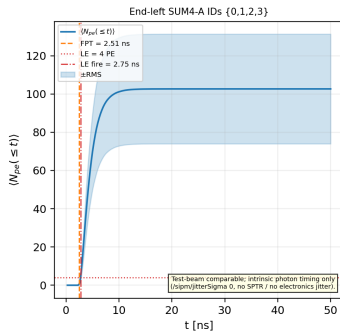
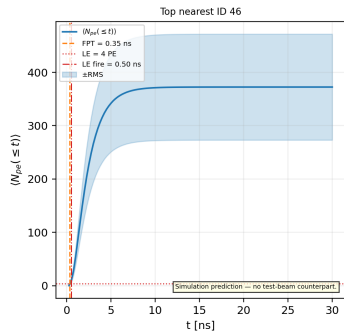
Position $x=-100$ mm



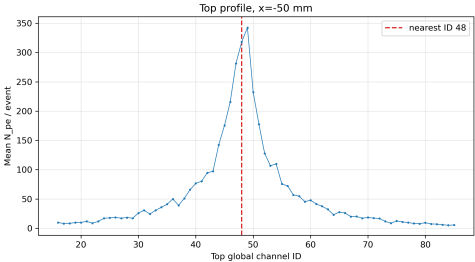
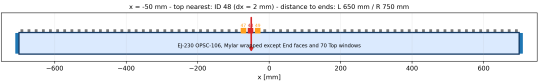
Quantity	Value
End L N_{pe}	203.9 ± 1.2
End R N_{pe}	115.5 ± 0.7
Nearest Top ID	46 (exact geometry)
Nearest Top N_{pe}	372.5 ± 2.2
Nearest Top $\langle t \rangle$	2.4913 ± 0.0019 ns
Nearest Top FPT	0.34565 ± 0.00016 ns

Run 14 | $x = -100$ mm | Photon arrival $\langle N(t) \rangle$

$x = -100$ mm | Photon arrival $\langle N(t) \rangle$



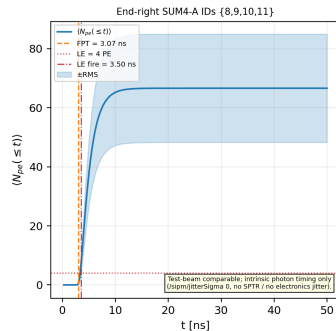
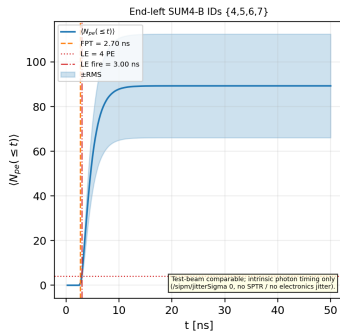
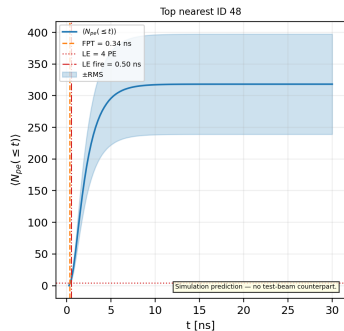
Position $x=-50$ mm



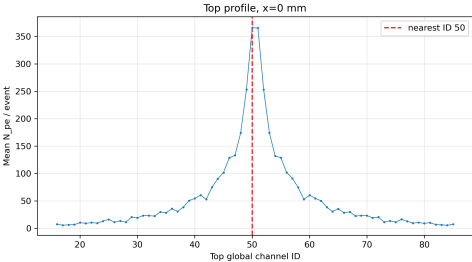
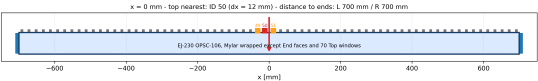
Quantity	Value
End L N_{pe}	176.1 ± 1.0
End R N_{pe}	131.0 ± 0.7
Nearest Top ID	48 (window-track exception)
Nearest Top N_{pe}	318.2 ± 1.8
Nearest Top $\langle t \rangle$	2.3422 ± 0.0020 ns
Nearest Top FPT	0.34234 ± 0.00017 ns

Run 15 | $x = -50$ mm | Photon arrival $\langle N(t) \rangle$

$x = -50$ mm | Photon arrival $\langle N(t) \rangle$



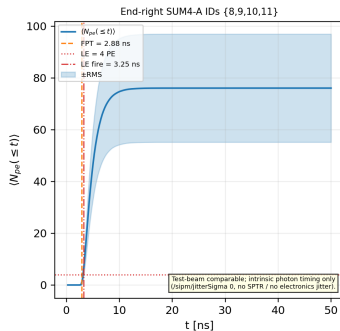
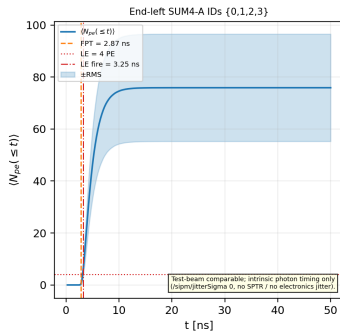
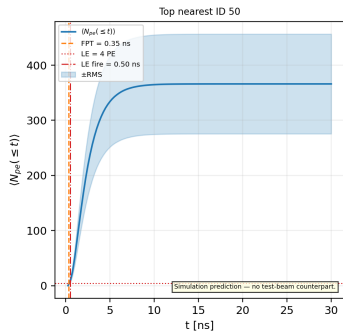
Position $x=0$ mm



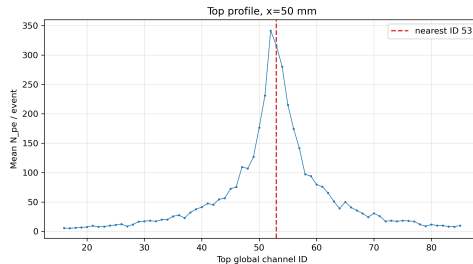
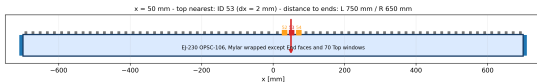
Quantity	Value
End L N_{pe}	151.0 ± 0.9
End R N_{pe}	151.1 ± 0.9
Nearest Top ID	50 (exact geometry)
Nearest Top N_{pe}	366.0 ± 2.0
Nearest Top $\langle t \rangle$	2.5528 ± 0.0020 ns
Nearest Top FPT	0.35099 ± 0.00023 ns

Run 16 | $x = +0$ mm | Photon arrival $\langle N(t) \rangle$

$x = +0$ mm | Photon arrival $\langle N(t) \rangle$



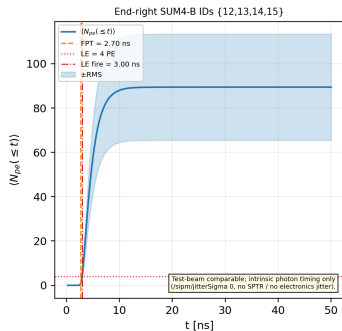
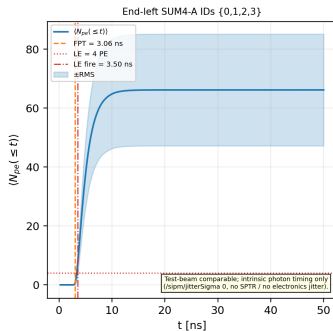
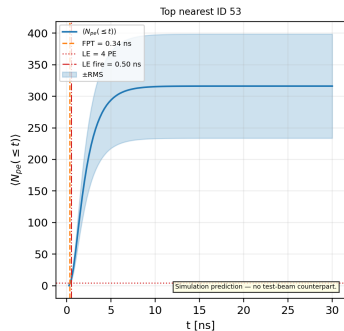
Position $x=50$ mm



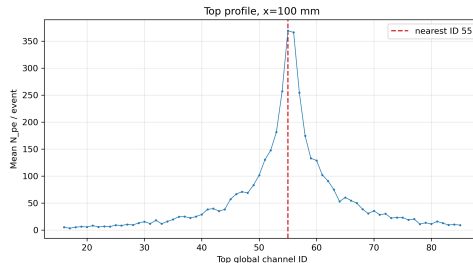
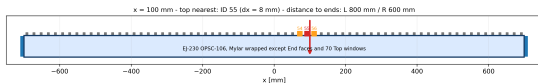
Quantity	Value
End L N_{pe}	129.9 ± 0.8
End R N_{pe}	175.9 ± 1.0
Nearest Top ID	53 (window-track exception)
Nearest Top N_{pe}	316.2 ± 1.8
Nearest Top $\langle t \rangle$	2.3413 ± 0.0020 ns
Nearest Top FPT	0.34218 ± 0.00014 ns

Run 17 | $x = +50$ mm | Photon arrival $\langle N(t) \rangle$

$x = +50$ mm | Photon arrival $\langle N(t) \rangle$



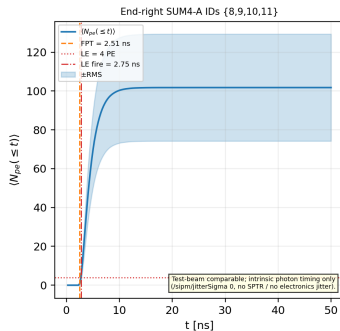
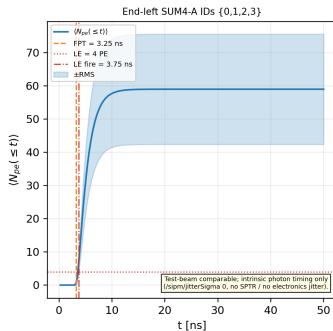
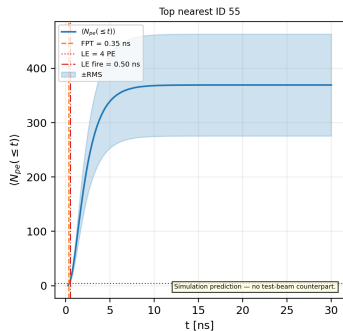
Position $x=100$ mm



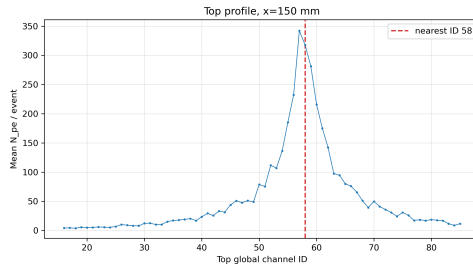
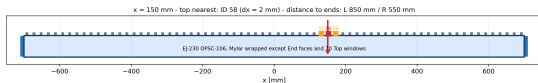
Quantity	Value
End L N_{pe}	114.4 ± 0.7
End R N_{pe}	202.3 ± 1.2
Nearest Top ID	55 (exact geometry)
Nearest Top N_{pe}	369.3 ± 2.1
Nearest Top $\langle t \rangle$	2.4904 ± 0.0019 ns
Nearest Top FPT	0.34584 ± 0.00017 ns

Run 18 | $x = +100$ mm | Photon arrival $\langle N(t) \rangle$

$x = +100$ mm | Photon arrival $\langle N(t) \rangle$



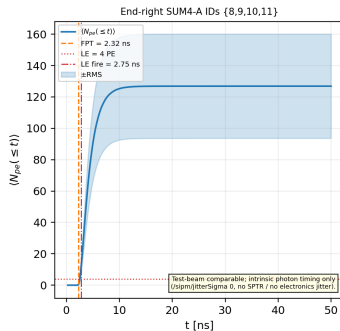
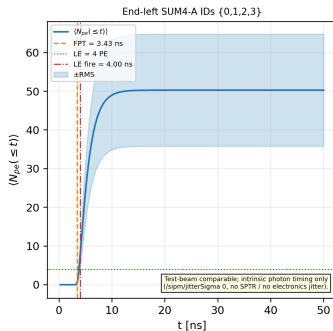
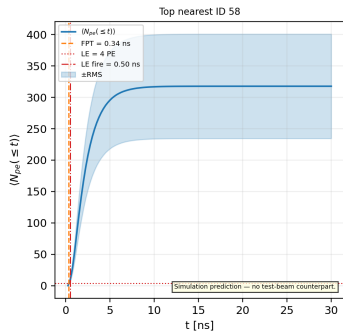
Position $x=150$ mm



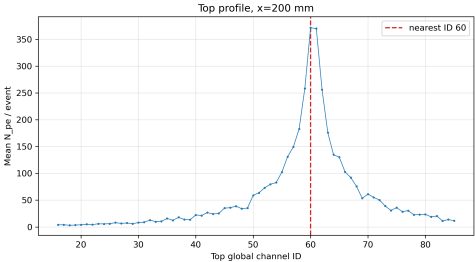
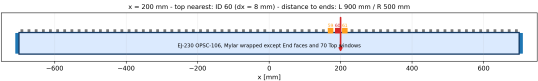
Quantity	Value
End L N_{pe}	100.5 ± 0.6
End R N_{pe}	240.4 ± 1.4
Nearest Top ID	58 (window-track exception)
Nearest Top N_{pe}	317.7 ± 1.9
Nearest Top $\langle t \rangle$	2.3405 ± 0.0020 ns
Nearest Top FPT	0.34244 ± 0.00018 ns

Run 19 | $x = +150$ mm | Photon arrival $\langle N(t) \rangle$

$x = +150$ mm | Photon arrival $\langle N(t) \rangle$



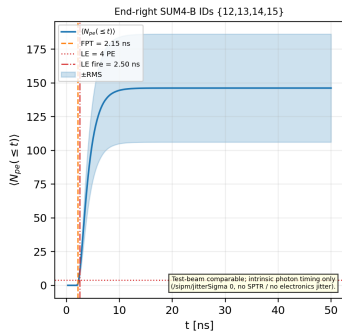
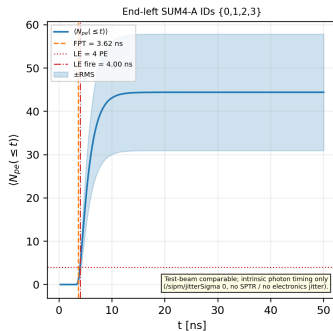
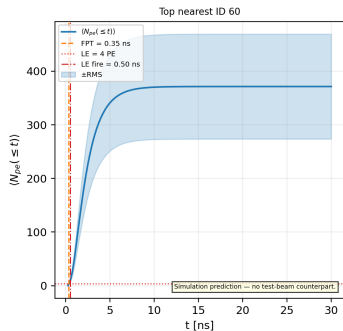
Position $x=200$ mm



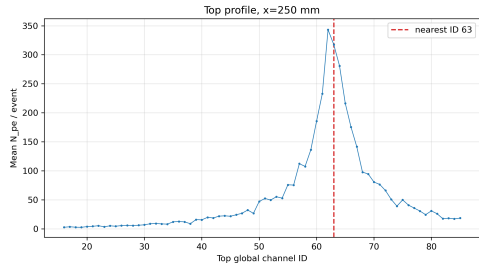
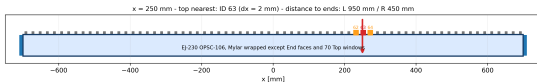
Quantity	Value
End L N_{pe}	88.0 ± 0.6
End R N_{pe}	285.6 ± 1.7
Nearest Top ID	60 (exact geometry)
Nearest Top N_{pe}	371.5 ± 2.2
Nearest Top $\langle t \rangle$	2.4939 ± 0.0019 ns
Nearest Top FPT	0.34568 ± 0.00021 ns

Run 20 | $x = +200$ mm | Photon arrival $\langle N(t) \rangle$

$x = +200$ mm | Photon arrival $\langle N(t) \rangle$



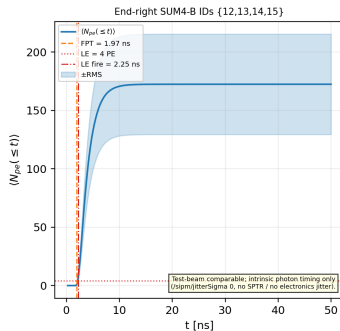
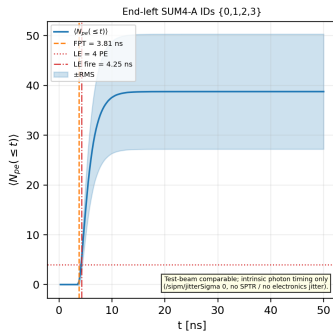
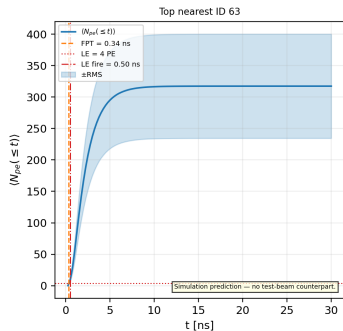
Position $x=250$ mm



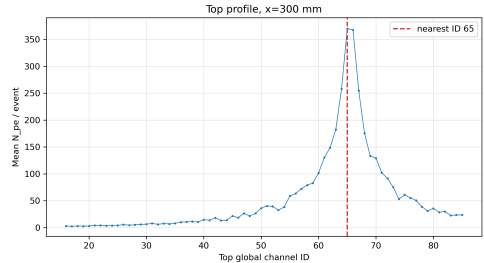
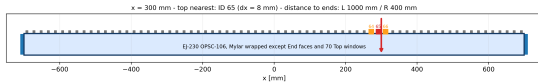
Quantity	Value
End L N_{pe}	76.7 ± 0.5
End R N_{pe}	337.3 ± 1.9
Nearest Top ID	63 (window-track exception)
Nearest Top N_{pe}	317.5 ± 1.9
Nearest Top $\langle t \rangle$	2.3414 ± 0.0020 ns
Nearest Top FPT	0.34205 ± 0.00013 ns

Run 21 | $x = +250$ mm | Photon arrival $\langle N(t) \rangle$

$x = +250$ mm | Photon arrival $\langle N(t) \rangle$



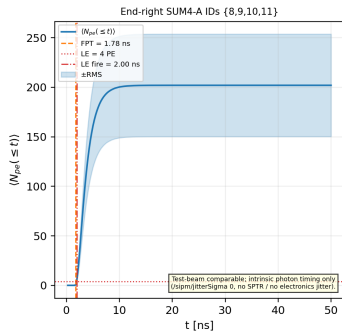
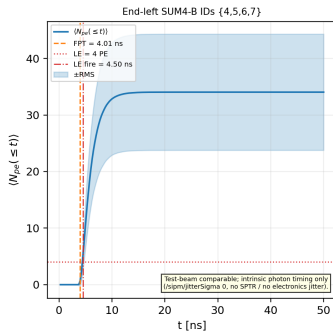
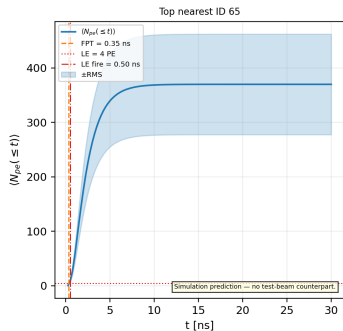
Position $x=300$ mm



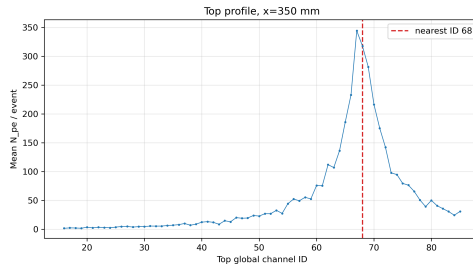
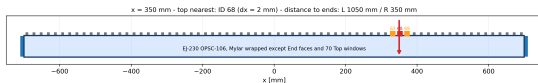
Quantity	Value
End L N_{pe}	67.4 ± 0.4
End R N_{pe}	396.4 ± 2.2
Nearest Top ID	65 (exact geometry)
Nearest Top N_{pe}	369.9 ± 2.1
Nearest Top $\langle t \rangle$	2.4957 ± 0.0020 ns
Nearest Top FPT	0.34550 ± 0.00014 ns

Run 22 | $x = +300$ mm | Photon arrival $\langle N(t) \rangle$

$x = +300$ mm | Photon arrival $\langle N(t) \rangle$



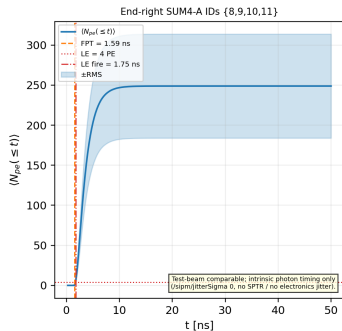
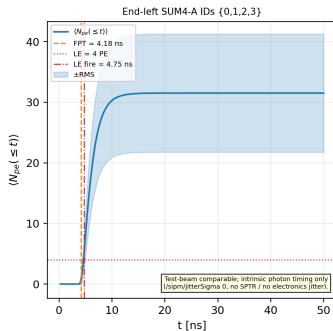
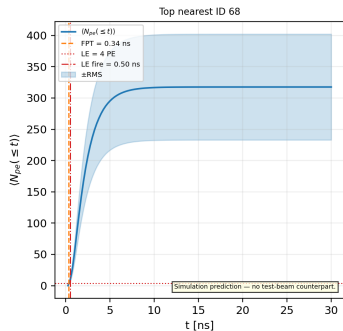
Position $x=350$ mm



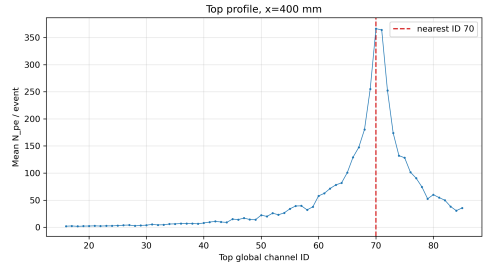
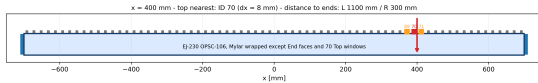
Quantity	Value
End L N_{pe}	60.6 ± 0.4
End R N_{pe}	480.0 ± 2.8
Nearest Top ID	68 (window-track exception)
Nearest Top N_{pe}	317.6 ± 1.9
Nearest Top $\langle t \rangle$	2.3425 ± 0.0020 ns
Nearest Top FPT	0.34251 ± 0.00021 ns

Run 23 | $x = +350$ mm | Photon arrival $\langle N(t) \rangle$

$x = +350$ mm | Photon arrival $\langle N(t) \rangle$



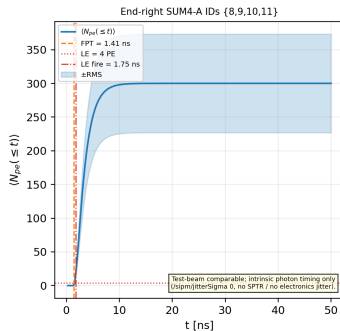
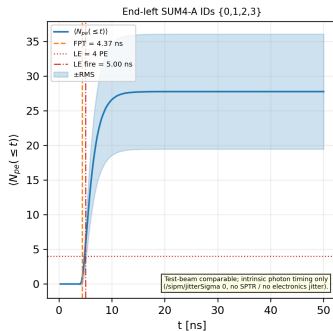
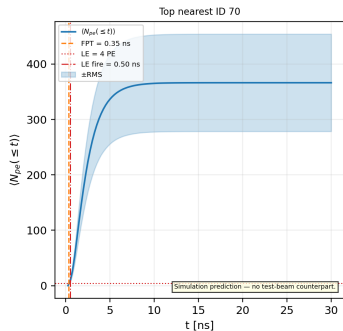
Position $x=400$ mm



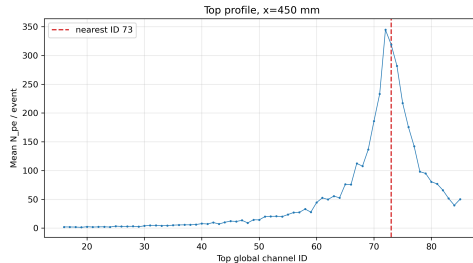
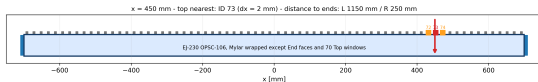
Quantity	Value
End L N_{pe}	52.8 ± 0.3
End R N_{pe}	577.8 ± 3.1
Nearest Top ID	70 (exact geometry)
Nearest Top N_{pe}	366.3 ± 2.0
Nearest Top $\langle t \rangle$	2.4909 ± 0.0020 ns
Nearest Top FPT	0.34615 ± 0.00025 ns

Run 24 | $x = +400$ mm | Photon arrival $\langle N(t) \rangle$

$x = +400$ mm | Photon arrival $\langle N(t) \rangle$



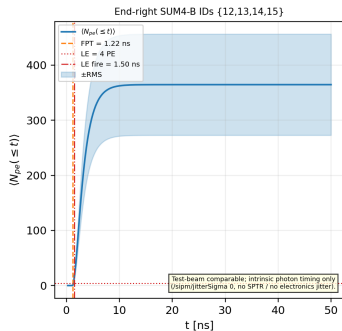
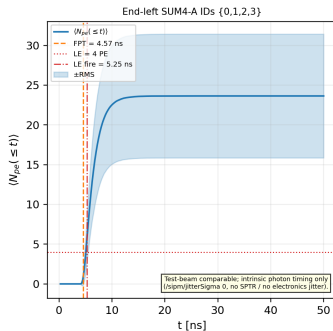
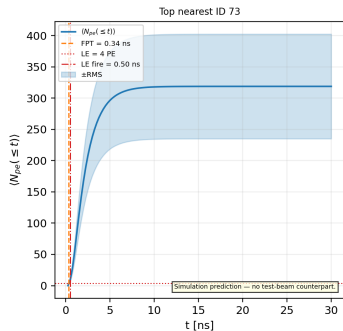
Position $x=450$ mm



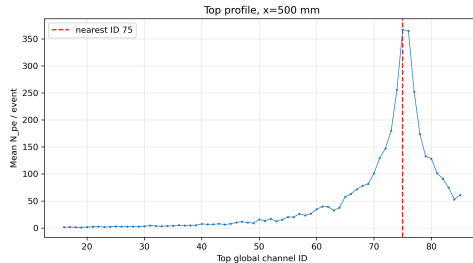
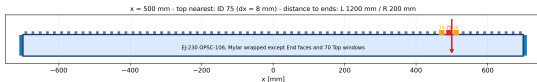
Quantity	Value
End L N_{pe}	47.0 ± 0.3
End R N_{pe}	719.4 ± 4.1
Nearest Top ID	73 (window-track exception)
Nearest Top N_{pe}	318.7 ± 1.9
Nearest Top $\langle t \rangle$	2.3429 ± 0.0020 ns
Nearest Top FPT	0.34232 ± 0.00013 ns

Run 25 | $x = +450$ mm | Photon arrival $\langle N(t) \rangle$

$x = +450$ mm | Photon arrival $\langle N(t) \rangle$



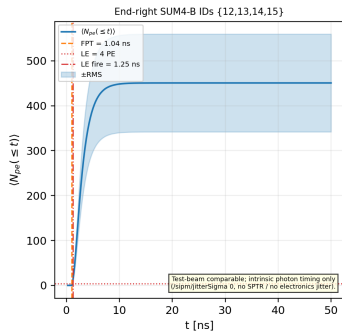
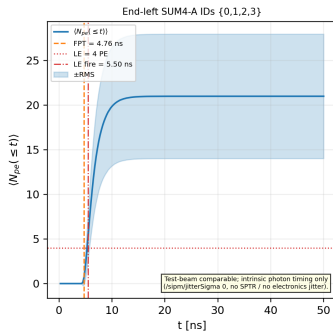
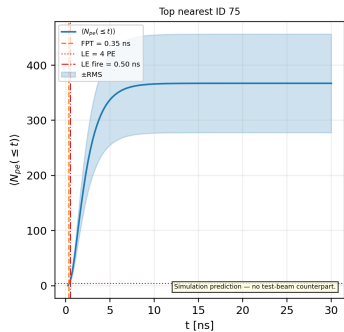
Position $x=500$ mm



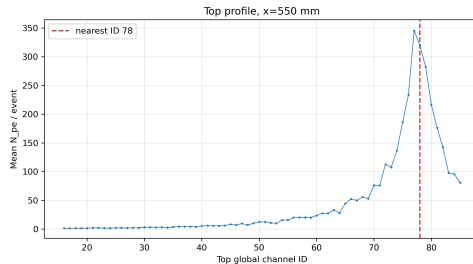
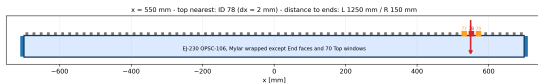
Quantity	Value
End L N_{pe}	41.6 ± 0.3
End R N_{pe}	890.1 ± 4.8
Nearest Top ID	75 (exact geometry)
Nearest Top N_{pe}	367.1 ± 2.0
Nearest Top $\langle t \rangle$	2.4910 ± 0.0020 ns
Nearest Top FPT	0.34605 ± 0.00020 ns

Run 26 | $x = +500$ mm | Photon arrival $\langle N(t) \rangle$

$x = +500$ mm | Photon arrival $\langle N(t) \rangle$



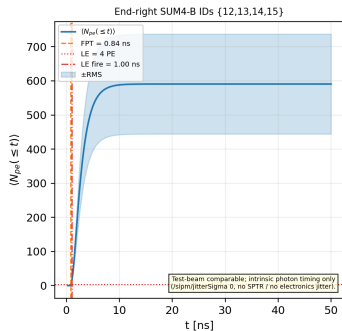
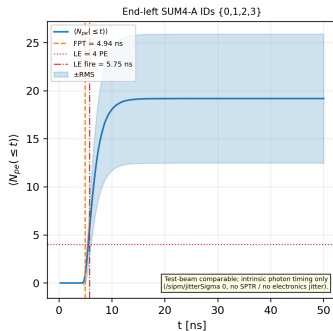
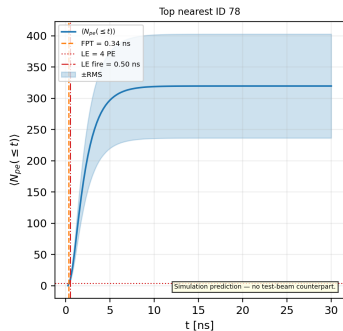
Position $x=550$ mm



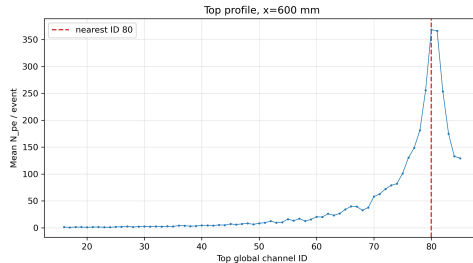
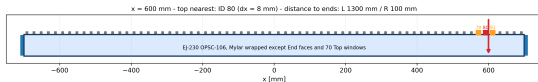
Quantity	Value
End L N_{pe}	37.8 ± 0.3
End R N_{pe}	1154.5 ± 6.5
Nearest Top ID	78 (window-track exception)
Nearest Top N_{pe}	319.6 ± 1.9
Nearest Top $\langle t \rangle$	2.3459 ± 0.0020 ns
Nearest Top FPT	0.34241 ± 0.00014 ns

Run 27 | $x = +550$ mm | Photon arrival $\langle N(t) \rangle$

$x = +550$ mm | Photon arrival $\langle N(t) \rangle$



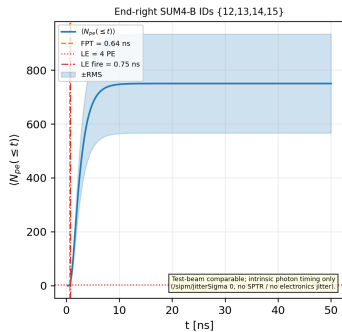
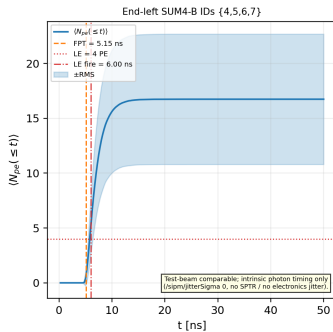
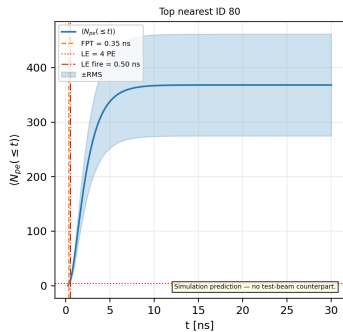
Position $x=600$ mm



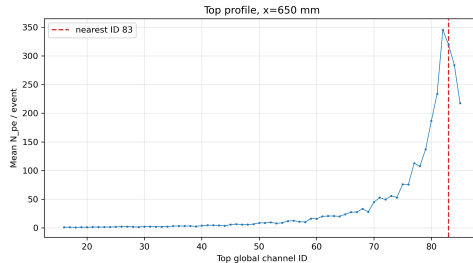
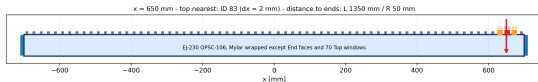
Quantity	Value
End L N_{pe}	33.1 ± 0.2
End R N_{pe}	1486.5 ± 8.1
Nearest Top ID	80 (exact geometry)
Nearest Top N_{pe}	368.2 ± 2.1
Nearest Top $\langle t \rangle$	2.4953 ± 0.0020 ns
Nearest Top FPT	0.34571 ± 0.00016 ns

Run 28 | $x = +600$ mm | Photon arrival $\langle N(t) \rangle$

$x = +600$ mm | Photon arrival $\langle N(t) \rangle$



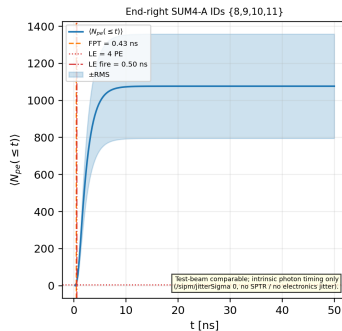
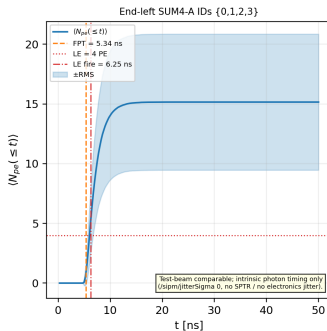
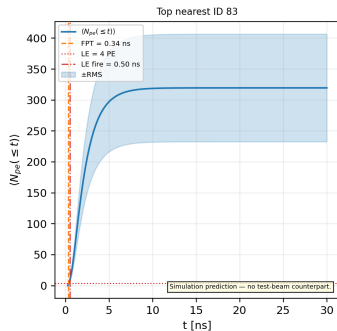
Position $x=650$ mm



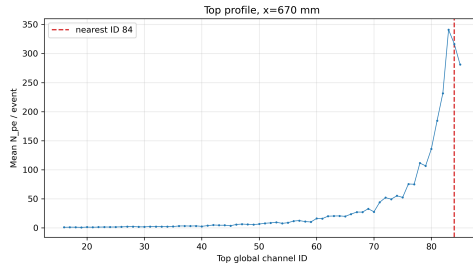
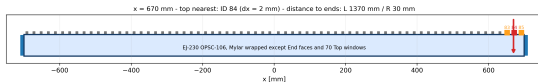
Quantity	Value
End L N_{pe}	30.2 ± 0.2
End R N_{pe}	2133.3 ± 12.4
Nearest Top ID	83 (window-track exception)
Nearest Top N_{pe}	319.7 ± 1.9
Nearest Top $\langle t \rangle$	2.3425 ± 0.0020 ns
Nearest Top FPT	0.34249 ± 0.00015 ns

Run 29 | $x = +650$ mm | Photon arrival $\langle N(t) \rangle$

$x = +650$ mm | Photon arrival $\langle N(t) \rangle$



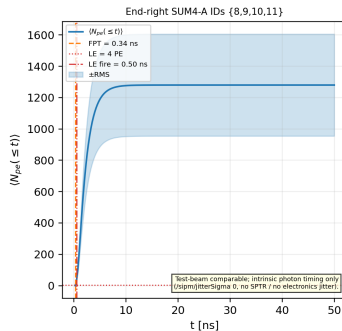
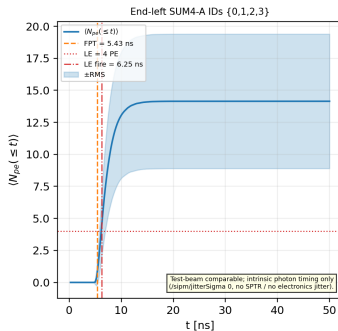
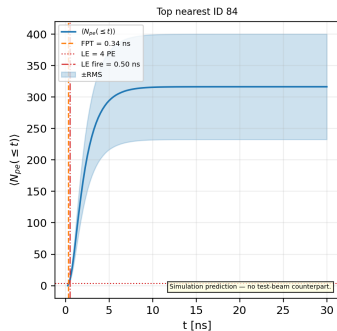
Position $x=670$ mm



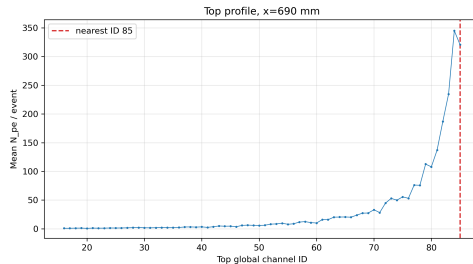
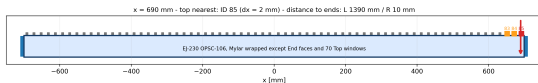
Quantity	Value
End L N_{pe}	28.2 ± 0.2
End R N_{pe}	2496.3 ± 14.1
Nearest Top ID	84 (window-track exception)
Nearest Top N_{pe}	316.4 ± 1.9
Nearest Top $\langle t \rangle$	2.3403 ± 0.0020 ns
Nearest Top FPT	0.34266 ± 0.00019 ns

Run 30 | $x = +670$ mm | Photon arrival $\langle N(t) \rangle$

$x = +670$ mm | Photon arrival $\langle N(t) \rangle$



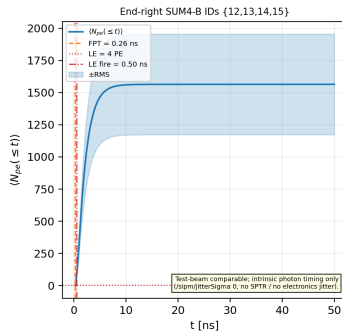
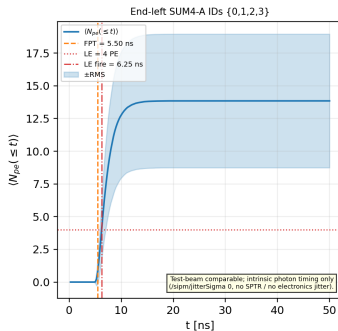
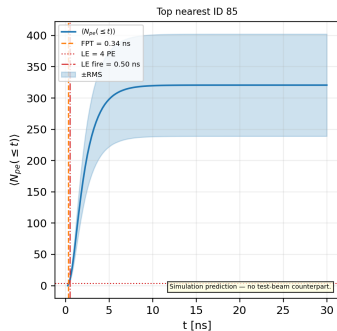
Position $x=690$ mm



Quantity	Value
End L N_{pe}	27.6 ± 0.2
End R N_{pe}	3089.6 ± 17.1
Nearest Top ID	85 (window-track exception)
Nearest Top N_{pe}	320.6 ± 1.8
Nearest Top $\langle t \rangle$	2.3455 ± 0.0020 ns
Nearest Top FPT	0.34224 ± 0.00015 ns

Run 31 | $x = +690$ mm | Photon arrival $\langle N(t) \rangle$

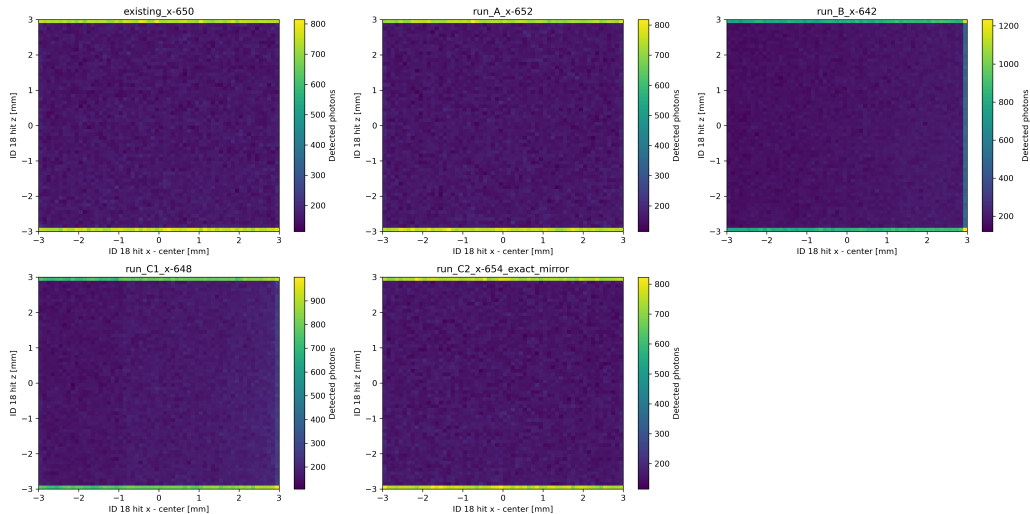
$x = +690$ mm | Photon arrival $\langle N(t) \rangle$



Backup: validation and gates

- ▶ 31/31 main ROOT files pass: event IDs 0–1999, matching beam x, aggregate IDs 0–85.
- ▶ Top localization passes all positions under the documented window-track/statistical-tie rules.
- ▶ At $x=300$ mm the two candidates differ by 0.00 sigma.
- ▶ MPT checks: RINDEX 1.58, yield 9700/MeV, decay 1.5 ns, rise 0.5 ns, ABSLENGTH 120 cm, emission peak ≈ 391 nm.

Backup: ID18 impact maps



Backup: why a 4-PE leading-edge threshold

1. **Noise rejection at hardware parity:** 4 PE is the nominal fixed threshold outside the correlated-noise spectrum.
2. **Slope optimum:** lower thresholds expose first-photon order statistics; higher thresholds reduce slope and increase walk.
3. **Far-end efficiency:** Minimum signal in the scan is End R 27.4 ± 0.2 at $x = -690$ mm; the nominal 4 PE threshold remains far below that signal.
4. **Walk:** no time-walk correction is applied in this campaign.

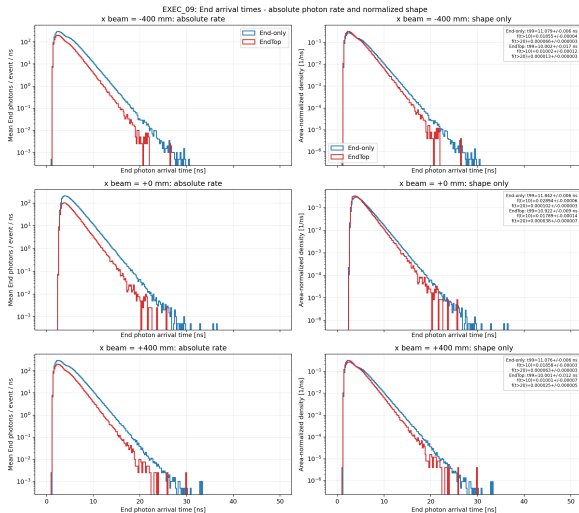
Backup: timing methodology

- ▶ SUM4 groups: $\{0..3\}$, $\{4..7\}$, $\{8..11\}$, $\{12..15\}$.
- ▶ Single-PE response: normalized 0.5/5 ns double exponential.
- ▶ Absolute leading-edge threshold: nominal 4 PE equivalent; the CSV-derived far-end rationale is reported on the preceding backup slide.
- ▶ Estimator: $\sigma(\Delta T_{AB})/\sqrt{2}$.
- ▶ No time-walk correction is applied in this campaign.
- ▶ The EndTop/End-only anti-artifact audit uses a common threshold, pulse, primary time convention and estimator.

Backup: glossary

FPT	First photon time: first detected photon per event.	SPTR	Single Photon Time Resolution: SiPM intrinsic single-PE jitter. Excluded from all EJ-230 results; all reported timing is intrinsic.
Fano factor	$F \equiv \text{Var}(N_{pe}) / \langle N_{pe} \rangle = 1 + c \langle N_{pe} \rangle$ when Landau ΔE fluctuations are present.	SUM4	Analog sum of four SiPMs: {0–3}, {4–7}, {8–11}, {12–15} (congruent_sum4_timing.C: 218–221).
Group A / Group B	Same-End SUM4 groups: A={0–3}, B={4–7} on End L; mirrors {8–11}, {12–15} on End R. $\sigma_{\text{group}} = \sigma(\Delta T_{AB}) / \sqrt{2}$ (sum4_timing.C:218–221).	Timing gate	exec08b_timing_gate.py tests EndTop vs. End-only σ_{group} at $x = 0, \pm 400$ mm; no time acceptance cut on photons.
Time-walk	No time-walk correction is applied in the EJ-230 campaign.	Window-track effect	Local symmetric ± 2 mm effect: 7.9% with significance $9.3\text{--}9.7\sigma$; centered/midpoint contrast significances: $-1.02\sigma / -0.06\sigma$.
Intrinsic sigma	Excludes SPTR and electronics jitter.	t_N / time-to-threshold	N -th detected photoelectron arrival time per event. $\sigma(t_N)$: photon-counting timing resolution at ideal N -PE digital threshold.
Moyal	Closed-form Landau approximation for stable MINUIT fits. Undershoots extreme tail.	(exact geometry)	Nearest SiPM = maximum, or window-track exception; $x = 300$ mm is a statistical tie.

Backup: EndTop versus End-only tail audit



Backup: far-End adaptive t_N (N_{eff} where 20 PE is unreachable)

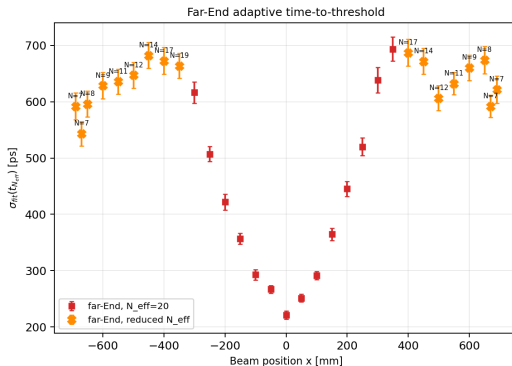
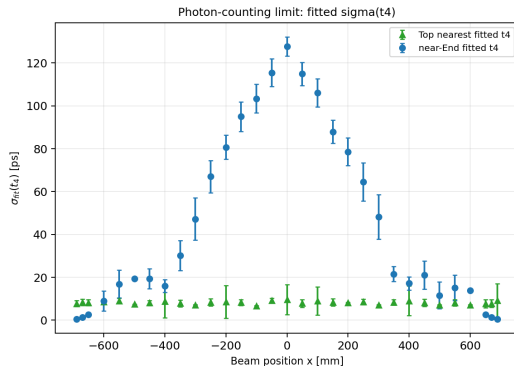
x	Top	near-End	far-End	$R_{far}(20)$
-690	4	4	7	10.8%
-670	4	4	7	14.0%
-650	4	4	8	16.4%
-600	4	4	9	25.1%
-550	4	4	11	41.9%
-500	4	4	12	54.4%
-450	4	4	14	69.2%
-400	4	4	17	88.3%
-350	4	4	19	94.0%
-300	4	4	20	98.0%
-250	4	4	20	99.6%
-200	4	4	20	100.0%
-150	4	4	20	100.0%
-100	4	4	20	100.0%
-50	4	4	20	100.0%
+0	4	4	20	100.0%

x	Top	near-End	far-End	$R_{far}(20)$
+50	4	4	20	100.0%
+100	4	4	20	100.0%
+150	4	4	20	100.0%
+200	4	4	20	99.9%
+250	4	4	20	99.7%
+300	4	4	20	97.9%
+350	4	4	20	95.5%
+400	4	4	17	88.5%
+450	4	4	14	70.5%
+500	4	4	12	52.6%
+550	4	4	11	41.6%
+600	4	4	9	25.6%
+650	4	4	8	16.9%
+670	4	4	7	12.6%
+690	4	4	7	11.9%

N_{eff} is the largest threshold with event reach at least 95%; nominal Top/near-End/far-End thresholds are 4/ 4/ 20 PE. No genuine-starvation point remains.

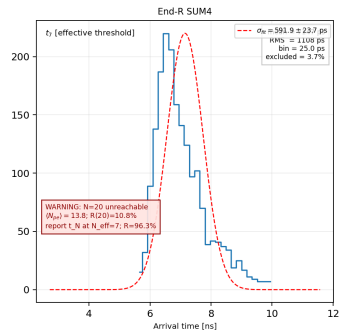
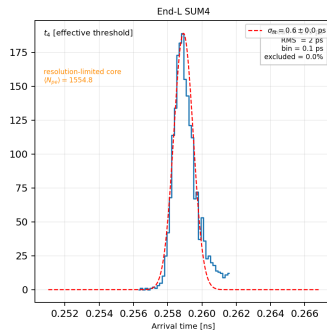
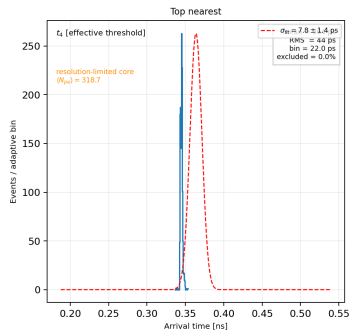
Backup: far-End adaptive t_N synthesis

EJ-230 fitted timing metrics with data-adaptive far-End threshold



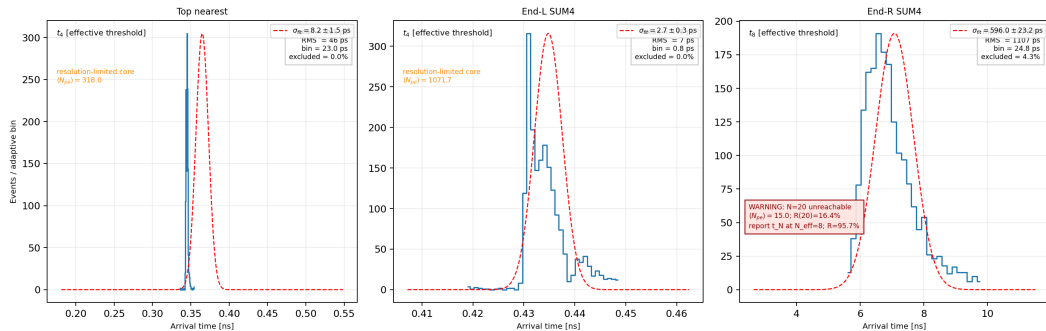
Backup: far-End adaptive t_N at $x = -690$ mm

EJ-230 t_N | $x = -690$ mm



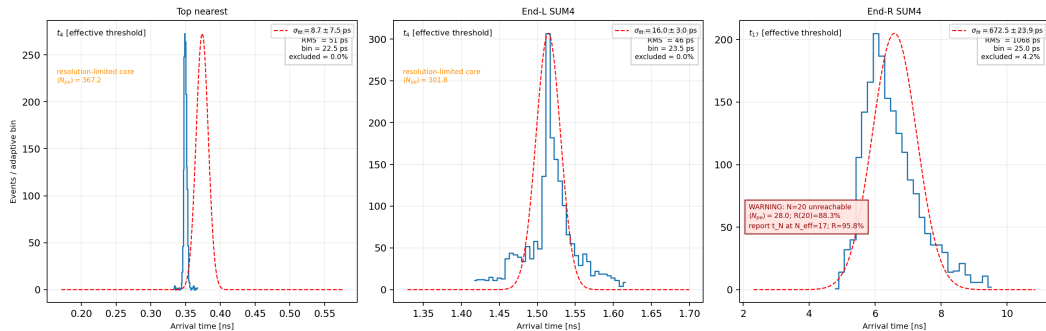
Backup: far-End adaptive t_N at $x = -650$ mm

EJ-230 t_N | $x = -650$ mm



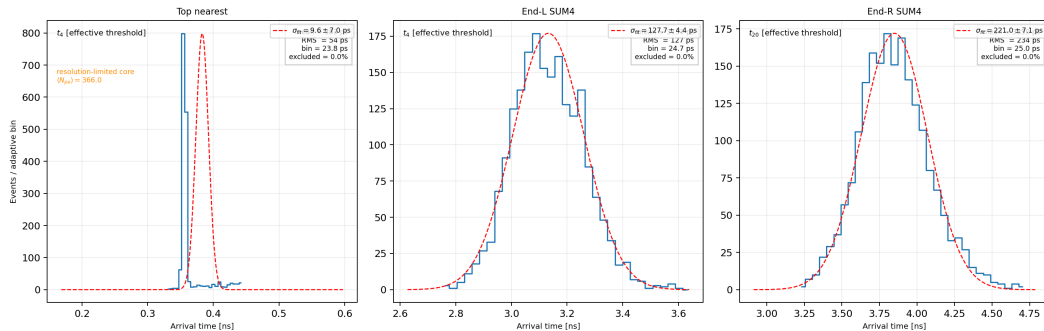
Backup: far-End adaptive t_N at $x = -400$ mm

EJ-230 t_N | $x = -400$ mm



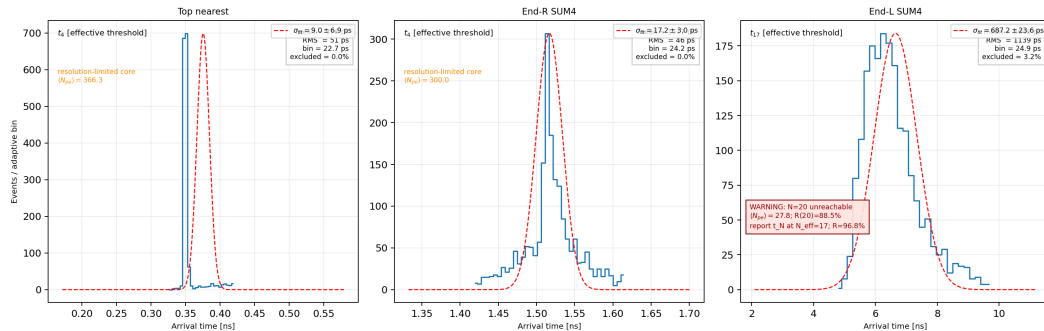
Backup: far-End adaptive t_N at $x = +0$ mm

EJ-230 t_N | $x = +0$ mm



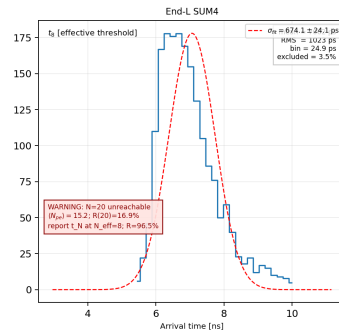
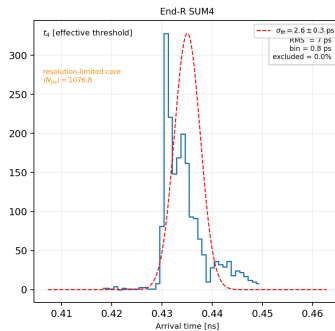
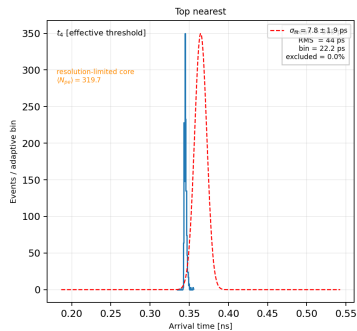
Backup: far-End adaptive t_N at $x = +400$ mm

EJ-230 t_N | $x = +400$ mm



Backup: far-End adaptive t_N at $x = +650$ mm

EJ-230 t_N | $x = +650$ mm



Backup: far-End adaptive t_N at $x = +690$ mm

EJ-230 t_N | $x = +690$ mm

